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2025 Engineering Design Package (EDP)



2025 Engineering Design Package (EDP)

University: California State University, Long Beach	Vehicle Name: Sharkus Electrus
Team Captain Name: Gustavo De la Torre	
Team Captain Email: gustavo.delatorregarcia01@student.csulb.edu	
Faculty Supervisor: Dr. Ji-Hwan Kang	Supervisor Email: jihwan.kang@csulb.edu
Revision #: 0	Revision Date: 03/07/2025
Onsite Safety Judge: (Virtual Only)	
Onsite Track Length: (Virtual Only)	*Onsite Track Width*: (Virtual Only)

Instructions for Chem-E-Car Team:

- All Chem-E-Car Teams competing in an AIChE Chem-E-Car Competition must complete and submit and Engineering Design Package by the posted deadline.
- Failure to meet the posted deadline will result in exclusion from the competition.
- Please complete all the applicable sections of this document, and save in PDF format. If your EDP is incomplete, AIChE cannot guarantee that it will be reviewed.
- Please rename the title of the document using the format University Name EDP.
- Additional information including Safety Data Sheets (SDS), Manufacturer's specification documents or specifications for custom-built components should be compiled and saved as another single PDF titled "University Name EDP Supplement".
- All teams will receive EDP Feedback with any suggested changes from AIChE Chem-E-Car Competition Safety Judges.
- Please use the MOC Form document to document changes made to the EDP after receiving EDP feedback from your online review.
- Each team must bring a printed copy of final EDP, EDP Supplement, EDP feedback and MOC Form in a folder or binder. This will help streamline the Onsite Safety Inspection on competition day.
- *Please review the AIChE [Code of Conduct](#) and [Code of Ethics](#).*

For more information, visit www.aiche.org/chemecar.

Job Safety Assessment Form

Car operation, hazards, and safety: Describe your Chem-E-Car's design, intended mode of operation (propulsion system), intended mode of control (stopping mechanism), auxiliary processes, major hazards and their control.

Describe your car's design: The car's battery system utilizes a dual-cell design, each incorporating five thermoelectric generators (TEGs). These TEGs are mounted on aluminum heat sinks, which in turn rest upon a 3D-printed cabinet designed to hold dry ice. Above the TEGs, a stainless-steel compartment will house the exothermic **reaction between hydrochloric acid (HCl) and a sodium hydroxide (NaOH) solution.**

Power source (propulsion system): Thermoelectric generators in series activated by a temperature difference between a cold source (dry ice) and a hot source (acid-base exothermic reaction). The TEGs are between the cold and hot sources and on top of a heat sink.

Auxiliary processes including synthesized fuel production, battery charging, hydrogen production, etc. (if applicable): N/A

Stopping mechanism: Our stopping mechanism employs the 'chemical chameleon' reaction, a visible chemical process where potassium permanganate and sugar-sodium hydroxide solutions are mixed. The resulting solution progresses through a distinct sequence of color changes. A strategically placed color sensor continuously monitors the reaction. When the solution reaches a pre-defined target color, such as green, the sensor will tell the motor to stop. This provides a precise, color-based stopping control.

Hazards inherent in design (for car & auxiliary processes, if applicable): Cryogenic burns and asphyxiation from dry ice, exothermic reactions and, oxidizing and irritating properties of potassium permanganate, corrosive and heat-generating NaOH, combustible sucrose, and electrical risks like short circuits and battery hazards.

Safety measures to prevent and mitigate hazards. Every hazard identified above must be addressed here: Using PPE and ventilation for dry ice and chemical handling, containing all reactions, insulating electrical components

Teams must implement a change in the reaction chemistry to both their propulsion and if applicable stopping reaction from the previous competition year.

Please list the major design changes for your car, and how the reaction chemistry of the propulsion and stopping mechanism (where applicable) has been changed from the car used in the previous year: The whole design has changed. Where last year we used an Aluminum-Air Battery as the propulsion system with an Iodine-clock stopping mechanism.

For cars & also for auxiliary processes if applicable

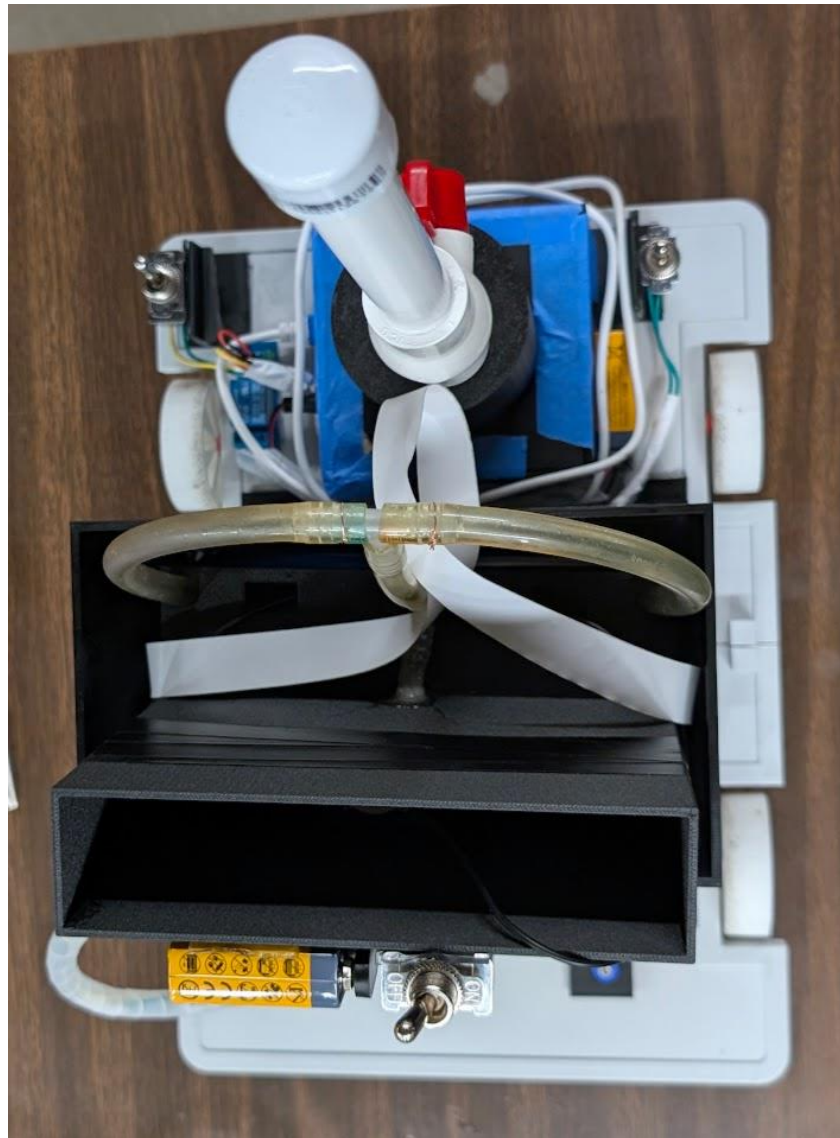
Expected Operating Conditions: Ambient
Temperature: 50 °C

If your car generates pressure above 5 psig, please list the Maximum Operating Pressure (MOP) and the Maximum Allowable Working Pressure (MAWP): N/A

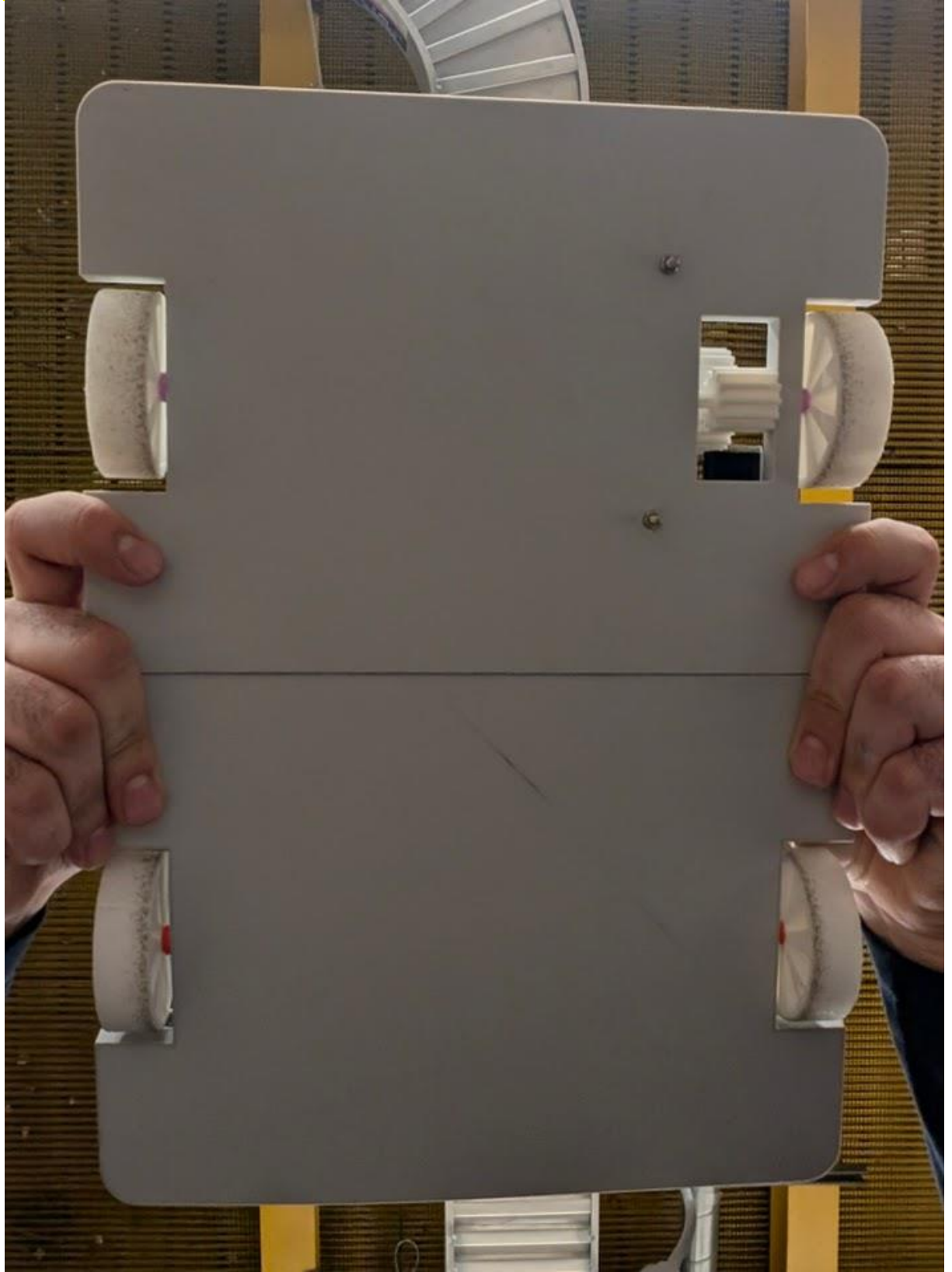
Photos of Completed Vehicle:

Please add photos of your vehicle after construction has been completed. These pictures must be current. The entire car must be visible in at least one picture. Remove the top to expose electrical controls if necessary. You must include multiple (at least 6) views of the car (**top, bottom, left, right, front and back**). Please adjust the following cells to fit *A drawing or AutoCAD document will NOT be accepted*

Top View



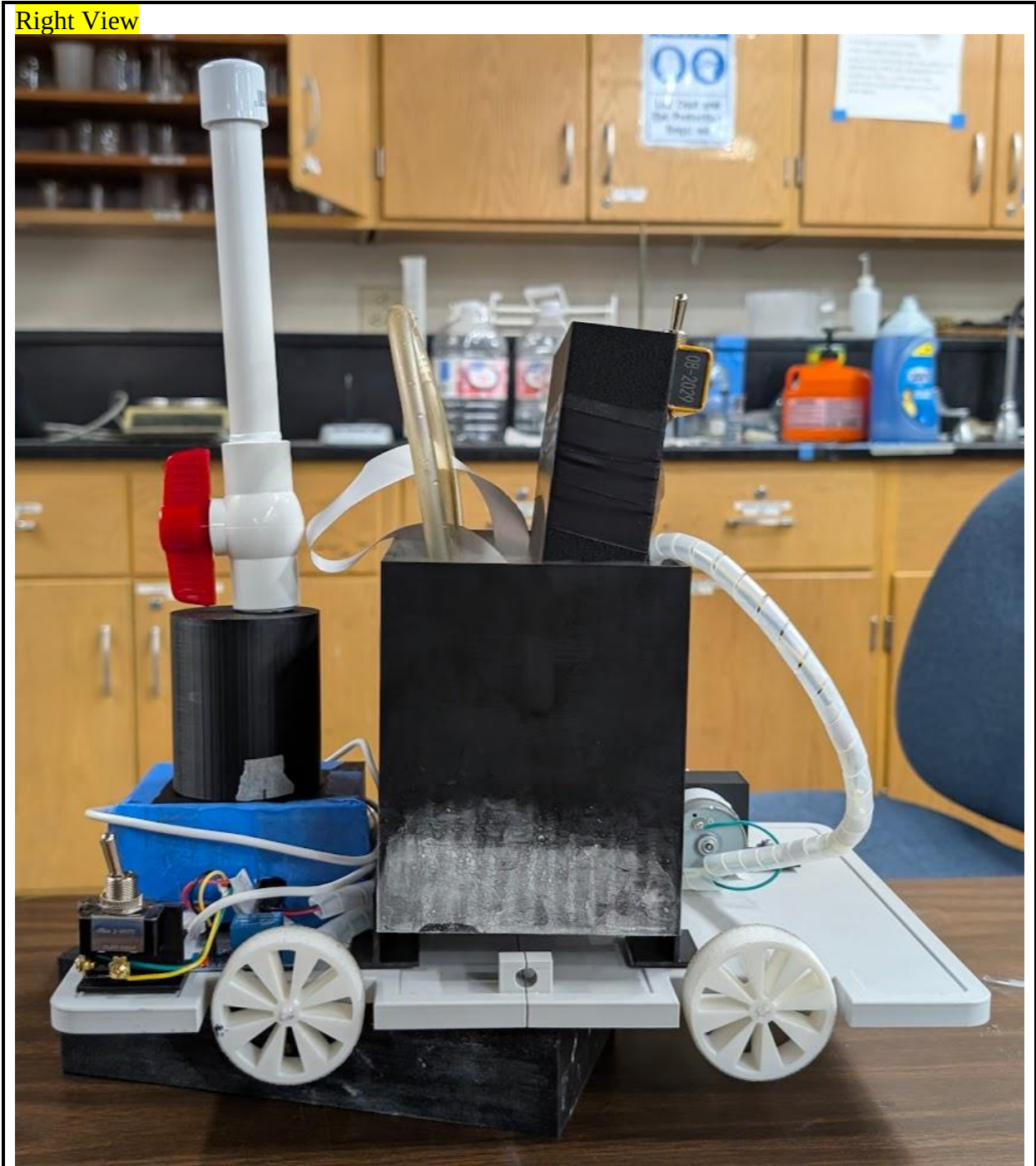
Bottom View



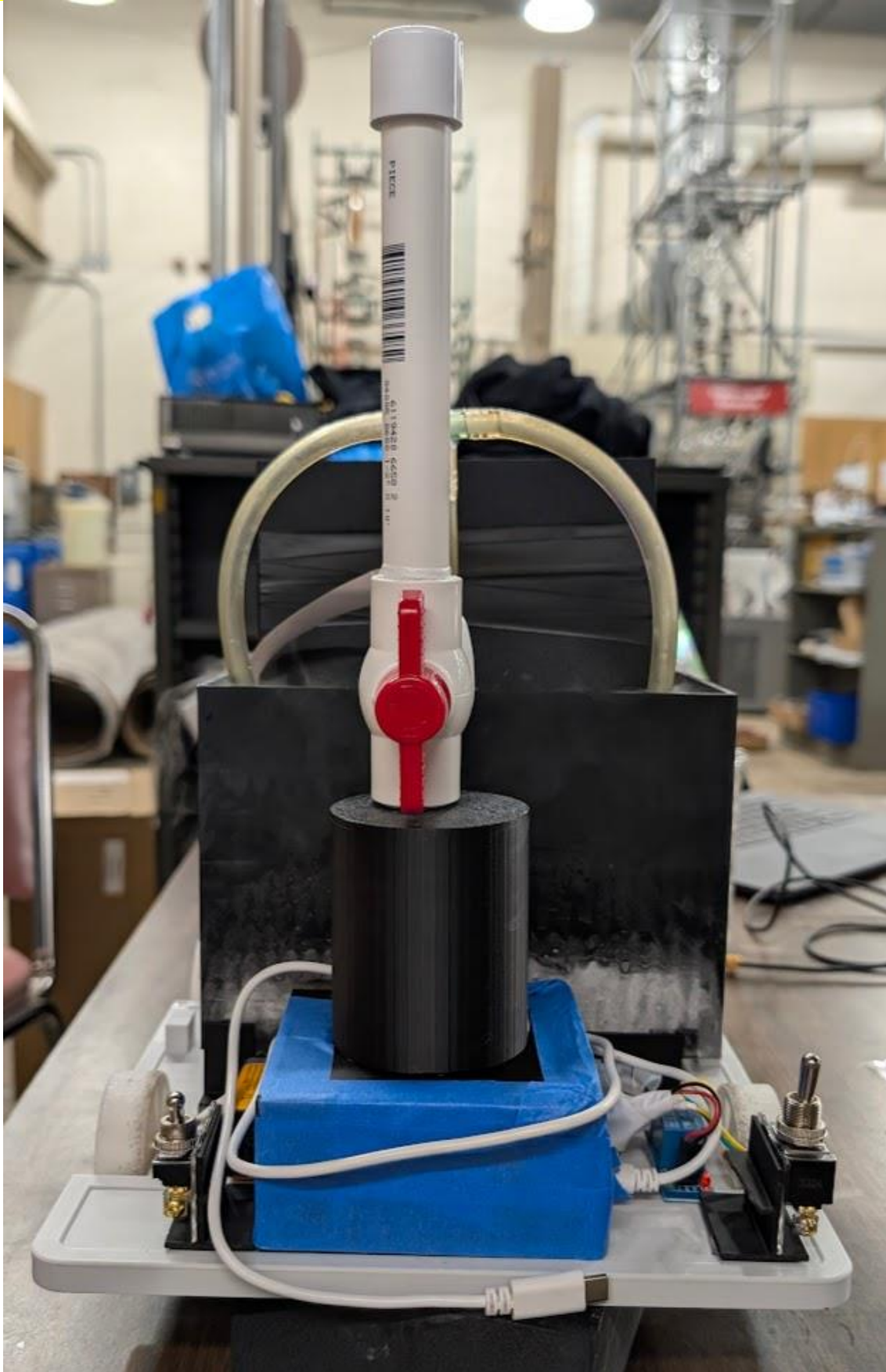
Left View



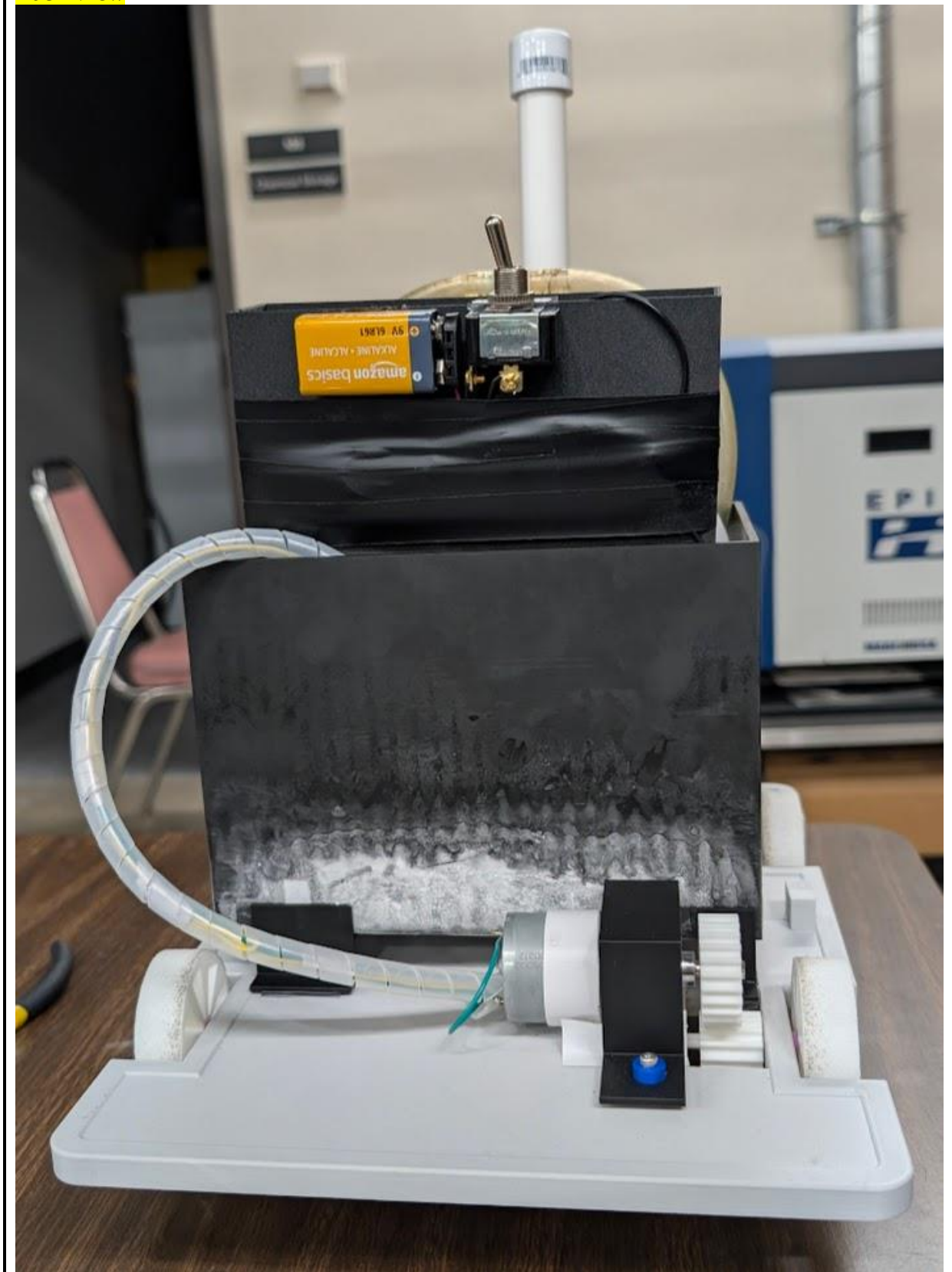
Right View



Front View



Back View



Request for Power Outlet at Competition Site (requirement for in-person competition): If your team requires a power outlet at your table at the competition, please provide reasoning here. *Please note that outlets requested to plug in laptops at your table will not be granted.*

N/A

Safety Training and Rules Certifications

University: California State University, Long Beach

Vehicle Name: Sharkus Electrus

1. Briefly describe the propulsion system (reaction/mechanism) of your vehicle.

Temperature difference between cold and hot sources creates energy in the thermoelectric generator.

2. Briefly describe the reaction that your vehicle uses to stop at the designated finish line.

A color sensor triggers a motor stop when a "chemical chameleon" reaction reaches a predetermined color.

3. Provide a precise answer to the following question: If your vehicle is 3 m short of the designated finish line on the first run, what specific changes will your team make to the stopping reaction to correct the shortfall?

We will try to decrease the concentration of the sodium hydroxide to decrease the time it takes for the solution to change to desired color.

4. Required Safety Training:

All participants must take and pass the AIChE Chem-E-Car Safety Training Course with a minimum score of 80%. For information on the Safety Training Course, please visit www.aiche.org/chemecar.

Please list the date that the required Chem-E-Car safety training that was completed by the faculty advisor:

Advisor Name	Date
<u>Dr. Ji-Hwan Kang</u>	<u>03/21/2025</u>

Please list the date the required Chem-E-Car safety training that was completed by each team member:

Team Member Name	Date
<u>Gustavo De la Torre</u>	<u>03/07/2025</u>
<u>Adrian Zentento</u>	<u>03/21/2025</u>
<u>Alejandro Jimenez</u>	<u>03/21/2025</u>
<u>Jason Aguilar</u>	<u>03/21/2025</u>
<u>Wailin Thai</u>	<u>03/21/2025</u>
<u>Isabel Valdez</u>	<u>03/21/2025</u>
<u>Mathew Tan</u>	<u>03/21/2025</u>
<u>Kevin Salazar</u>	<u>03/21/2025</u>
<u>Taylor Soop</u>	<u>10/27/2024</u>

5. Faculty Safety Rules Certification:

I certify that this student team has followed all of the safety and competition rules, has completed an engineering documentation package, has completed a safety review under my supervision or with an outside expert, and has at least ten hours of operating experience beyond the time required to design and assemble the car:

Faculty Advisor Name Ji-Hwan Kang

Faculty Advisor Signature Ji-Hwan Kang Date 03/21/2025

Outside Expert Name* _____

Outside Expert Signature* _____ Date _____

**Note that this section must be signed by (1) Chem E Car Faculty advisor at a minimum. Having a signature from another outside expert is not required but recommended if you are getting support from another faculty member or safety professional on this project.*

6. Student Safety Rules Certification:

We certify that we have followed all of the safety and competition rules, have completed an engineering documentation package, have completed a safety review with our faculty supervisor or with an outside expert, and have at least ten hours of operating experience beyond the time required to design and assemble the car. We understand and agree that we will not be allowed to compete in the Chem-E-Car Competition if our completed EDP package is not resubmitted by the posted deadline:

Team Member Signature

Date

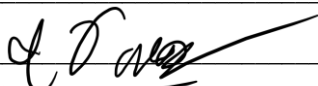




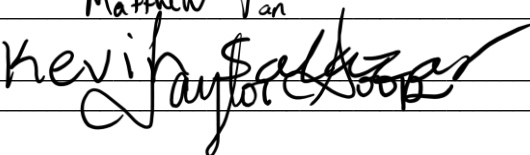


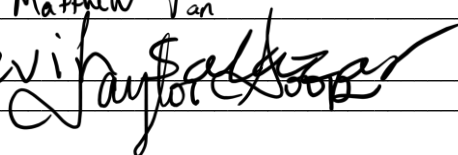
Jason Aguilar

Walter



Matthew Tan





03/17/2025

03/17/2025

03/17/2025

03/17/2025

03/17/2025

03/17/2025

03/17/2025

03/17/2025

03/20/2025

****Please include a copy of the Safety Training Course Certificate for each Team Member + Advisor in your EDP Supplement Document; Certificates should have a completion date of on or after January 1st 2025****



This Certificate of Completion certifies that
Ji-Hwan Kang
Has successfully completed:
ELA199 ChemE Car Safety Course

Date Awarded:
March 21, 2025

Rukyah Hennessey

Rukyah Hennessey
Director, Education Services Department



This Certificate of Completion certifies that
Gustavo De La Torre
Has successfully completed:
ELA199 ChemE Car Safety Course

Date Awarded:
March 7, 2025

Rukyah Hennessey

Rukyah Hennessey
Director, Education Services Department



This Certificate of Completion certifies that

Adrian Zenteno

Has successfully completed:

ELA199 ChemE Car Safety Course

Date Awarded:

March 21, 2025

Rukyah Hennessey

Rukyah Hennessey
Director, Education Services Department



This Certificate of Completion certifies that

Alejandro Jimenez

Has successfully completed:

ELA199 ChemE Car Safety Course

Date Awarded:

March 21, 2025

Rukyah Hennessey

Rukyah Hennessey
Director, Education Services Department



This Certificate of Completion certifies that
Jason De Jesus Aguilar Ramirez
Has successfully completed:
ELA199 ChemE Car Safety Course

Date Awarded:
March 21, 2025

Rukyah Hennessey

Rukyah Hennessey
Director, Education Services Department



This Certificate of Completion certifies that
Wailin Thai
Has successfully completed:
ELA199 ChemE Car Safety Course

Date Awarded:
March 21, 2025

Rukyah Hennessey

Rukyah Hennessey
Director, Education Services Department



This Certificate of Completion certifies that
Isabel Valdez
Has successfully completed:
ELA199 ChemE Car Safety Course

Date Awarded:
March 21, 2025

Rukyah Hennessey

Rukyah Hennessey
Director, Education Services Department



This Certificate of Completion certifies that
Matthew Tan
Has successfully completed:
ELA199 ChemE Car Safety Course

Date Awarded:
March 21, 2025

Rukyah Hennessey

Rukyah Hennessey
Director, Education Services Department



This Certificate of Completion certifies that
Kevin Salazar Flores
Has successfully completed:
ELA199 ChemE Car Safety Course

Date Awarded:
March 21, 2025

Rukyah Hennessey

Rukyah Hennessey
Director, Education Services Department



This Certificate of Completion certifies that
Taylor Soop
Has successfully completed:
ELA199 ChemE Car Safety Course

Date Awarded:
October 27, 2024

Darlene Schuster

Darlene Schuster, Executive Director & CEO, AICHE®

Hazards Analysis

This analysis is for your home institution, not the competition site. Please attach a floor diagram of the laboratory where you will be building and testing your vehicle on the following page. List the location of available safety equipment and spill response supplies on this diagram.

Expected Operating Conditions:

Temperature	Pressure
Normal: Ambient	Normal: Ambient
Minimum: -78.5°C	Minimum: Ambient
Maximum: 102°C	Maximum: Ambient

Personal Protective Equipment (PPE): Check all PPE worn during operation of this Chem-E-Car. Do not list these in the procedure section.

☒ Long Pants	☒ Safety Glasses	☐ Hard Hat	☒ Apron
☒ Long Sleeves	☒ Splash Goggles	☒ Insulated Gloves	☐ Ear Protection
☒ Non-porous Shoes	☒ Face Shield	☒ Chemical Gloves	☐ Other:

Available Safety Equipment – Provide the location of each item shown below at your home institution where your vehicle will be operated and tested. Show the location of this equipment on your provided floor plan. **Answer all questions.** If not available, type “NA” in the field.

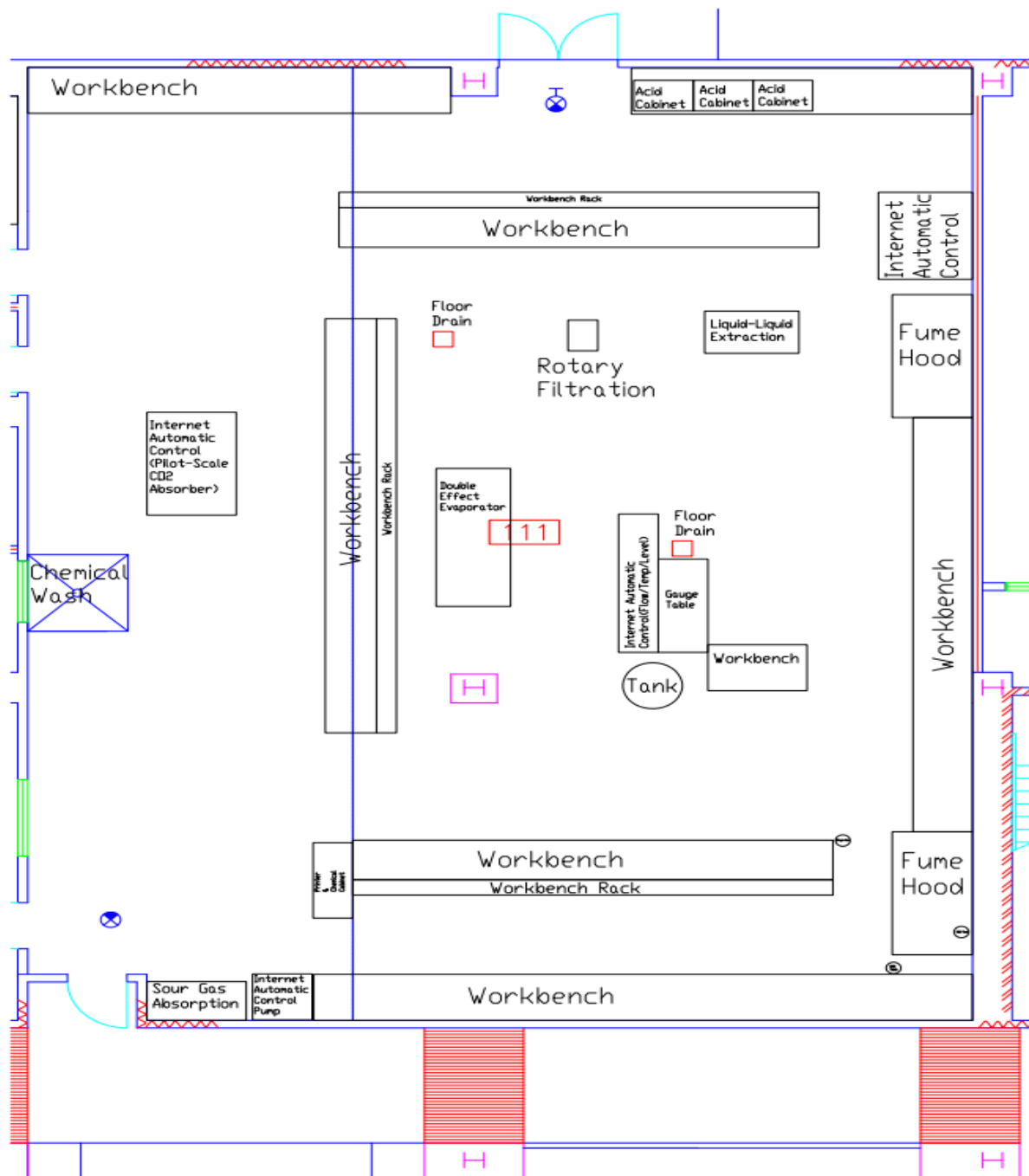
Item	Location
Fire Extinguisher:	ECS Room 111 against the North Wall
Eyewash:	ECS Room 111 against the North Wall
Safety Shower:	ECS Room 111 against the North Wall
Telephone:	ECS Room 111 against the West Wall
First Aid Kit:	ECS Room 111 against the West Wall
Spill Containment	N/A
Other:	

Spill Response Supplies - Provide the location of each item shown below at your home institution where your vehicle will be operated and tested. Show the location of this equipment on the attached floor plan. **Answer all questions.** If not available, type “NA” in the field.

Item	Location
Spill Kit:	ECS Room 111 Safety Cabinet
Sorbent Powder:	N/A
Spill Dikes:	N/A
Acid and/or Base Neutralizers:	ECS Room 111 Safety Cabinet
Drain Plugs:	ECS Room 111 Safety Cabinet
Spill Pillows:	N/A
Mercury Spill Kit:	N/A

Other:	
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Laboratory Floor Plan/Diagram: Please insert a floor plan diagram of the laboratory where you will be building and testing your vehicle on this page. List the location of available safety equipment and spill response supplies from the previous page on this diagram. Onsite Competition Fire & Safety Floor Plan showing the location of available fire and safety equipment as well as emergency exits in the performance competition venue will be provided to participating teams by the competition host.



Vehicle Primary Hazards Checklist: Check the box in the left hand column if the hazards listed below exist on the vehicle. Then check the applicable means of control for each hazard.

Hazard (check if present)	Control
☐ (a) Pressure	Anything greater than 5 psig. Must meet all requirements below: ☐ Pressure gauge (must read to 2x max. operating pressure) ☐ Emergency relief device set to no more than 1.1 times the max. operating pressure. Relief sizing calculations must be provided. ☐ Emergency relief device in proper location. ☐ Pressure certification – see Pressure Vessel Testing Protocol ☐ Proper management system to prevent over or mis-charging. ☐ All car components exposed to pressure must be certified to operate at that pressure. Provide manufacturer's pressure specifications. ☐ No PVC, cPVC or polyethylene terephthalate (PETE or PET) plastics in pressure service or at all Must have measurements or calculations to prove maximum operating pressure. Max allowable pressure this year is 200 psig. See ChemE car rules for more details on these requirements.
☒ (b) Hazardous Materials	Are any chemicals with a GHS hazard rating present (i.e., toxicity, flammability, corrosivity)? ☒ Doubly contained and handled properly. ☒ Team has properly filled out Team Waste Tags
☐ (c) Flammable Gasses	Are flammable gasses/vapors used in the operation of the car? ☐ Components exposed to flammable gas (i.e. fuel cells) are purged with an inert gas prior to use. Must be detailed in operating procedures. ☐ No tied balloons are allowed (i.e. helium balloons)
☒ (d) Temperature	Any exposed surface greater than 150 deg. F or under 32 deg F. ☒ Insulation or barrier to prevent contact.
☒ (e) Electrical	Exposed wiring and electrically energized components are ignition, electrocution, and a shorting/fire hazard. Alligator clips and twisted wire connections are not allowed; use binding posts or banana plugs for a more secure connection. ☒ Proper electrical insulation and connections provided. ☒ Wires are neat and orderly to prevent tangles/snags
☐ (f) Electro Chemical (Battery)	Any student fabricated electrochemical devices (batteries) used on the car? ☐ Electrochemical device to be fully disassembled into original components and individual components made safe (pH=6-8) and disposed of separately. Team shall explain process in detail. ☐ Team has properly filled out Team Waste Tags
☒ (g) Mechanical	Any parts (meshing gears, belts or chains) that are pinch hazards. ☒ Guards present and adequate.
☐ (h) Oxygen	All components exposed to oxygen. These must be ☐ certified for oxygen service. ☐ thoroughly cleaned of contaminants as per instructions in rules. ☐ not used previously for other types of service.
☐ (i) Biohazards	☐ No biohazards that require handling in a laboratory greater than Biosafety Level 1 (BSL-1) are permitted biohazard level 1 either during the design, development, preparation, or competition phases of your car.

Additional Fabrication & Operation Hazard Detail Check List: Check all hazards that are likely to be encountered during your Chem-Car construction and operation. List the major source(s) of the hazard and describe how the hazard(s) will be controlled. If both construction and hazard columns are checked in an individual row, then the hazards should be identified separately for both the construction and operation.

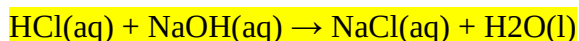
Hazard	Present During		Control Method(s) ¹	PPE Required ¹
	Construction?	Operation?		
Pressure	☐	☐		
Hazardous materials	☒	☒	Prevent skin contact with the reactions involved	Gloves, lab coat, and safety goggles during construction and operation
Hot Surfaces/High Temp > 150 F (65 C)	☐	☒	Prevents burns	Insulated heat gloves, lab coat, and safety goggles
Cold Surfaces/Low Temp < 32 F (0 C)	☐	☒	Prevents frostbite	Insulated gloves, lab coat, and safety goggles
Electrical	☒	☒	Wires are covered in insulation and capped	Gloves, lab coat, and safety goggles
Arc welding	☐	☐		
Gas welding	☐	☐		
Lathe	☐	☐		
Milling machine	☐	☐		
Handheld power tools	☐	☐		
Drill press	☐	☐		
Other mechanical hazards	☐	☐		
Paint spraying	☐	☐		
Ionizing radiation	☐	☐		
Laser radiation	☐	☐		
Asphyxiates	☐	☐		
Open flames	☐	☐		
Potential Spills	☒	☒	Reactions are covered and monitored	Gloves, lab coat, and safety goggles
Biohazards:	☐	☐		
Other:	☐	☐		
Other:	☐	☐		

Chemical Information Page

Description of Chemistry/Chemical Reactions: Provide details below on any chemical reaction(s) that occur in your system. Please show the reactants involved, the stoichiometry and the heat of reaction, if available. Also list side reactions and any other reactions that may impact safety.

Propulsion System:

- Thermoelectric generators (TEG's) create energy by a temperature difference in either one of its sides.
- Cold temperature comes from the dry ice
- Hot temperature comes from the reaction of:



- Where the water product from the first reaction is used in the second reaction to create a higher exothermic reaction

Stopping Mechanism:

- The stopping mechanism employs the "chemical chameleon" reaction, which utilizes the redox reactions of potassium permanganate (KMnO_4) in an alkaline solution to produce a series of color changes.
$$\text{MnO}_4^- \text{ (purple, Mn + 7)} \rightarrow \text{MnO}_4^{2-} \text{ (green, Mn + 6)} \rightarrow \text{MnO}_2 \text{ (brown, Mn + 4)}$$
- A reducing agent (like sugar) is added to a permanganate solution. The amount of reducing agent is carefully controlled to trigger the sequential reduction of permanganate, resulting in predictable color changes over time.
- A color sensor (such as a photodiode or RGB sensor) is used to monitor the changing colors of the solution. The sensor is calibrated to detect a specific color in the sequence (e.g., the transition to manganese dioxide, which is brownish-yellow).
- When the sensor detects the target color, it sends a signal to a microcontroller or circuit, which then cuts off the power to the car's motor, causing it to stop.

Table 1: Please list all chemicals, concentrations and quantities that will be used at the competition site. Concentration **MUST** be that of the raw material being shipped, not your race day solutions. Select the check-box for any shipped chemicals. This is so the Host can prepare to receive, store and transport your chemicals.

Chemical Name	Chemical State (Solid/Liq./Gas)	Concentration/ Units	Amount (units)	Shipped to Host School (Yes/No)
Dry ice (CO ₂)	Solid	45.5 M	600 g	N/A
Hydrochloric acid (HCl)	Liquid	3 M	300 mL	N/A
Potassium Permanganate (KMnO ₄)	Solid	.001 M	0.19 g	N/A

Note: Please include any special storage requests that the Host should be aware of for your chemicals. Note these requests must be coordinated with the host school directly, stating them here is not sufficient.

No chemicals needed to ship since we are hosting.

Table 2: Please list all chemicals that you expect to generate and dispose of during the competition. This should be WASTE/USED chemicals only. Chemicals must be made safe by the team for disposal in one of the listed waste classifications on the Team Waste Tag form. This section should exactly match the information on your waste tags. Team Waste Tags must be present to race and completed to be allowed to dispose of chemicals. See Team Waste Tags section of EDP.

Waste Description	Concentration (Include units!)	Amount (MUST include units!)	Disposal Waste Stream* (Flammable, Acid, Aqueous [pH 6-9], Organic, Base, Solids [pH 6-9], Other*) *Only 1 classification may be selected **If Other, classification must be specified
Dry ice (CO ₂)	45.5 M	300 g	Other
Hydrochloric acid (HCl)	3 M	300 mL	Acid
Sodium hydroxide (NaOH)	3 M	11 g	Base
Potassium Permanganate (KMnO ₄)	.001 M	.19 g	Base

Chemical Hazards and Disposal

Chemical Properties and Hazards for ALL CHEMICALS, including reactants, intermediates, products, and lubricants.

Chemical Name	Physical State (S, L, G)	GHS Hazard Classifications*	Incompatible Chemicals List chemicals present within the laboratory	Flash Point Temp.	Flammability Limits	
					LFL	UFL
Dry ice	S	May cause frostbite. May cause cryogenic burns or injury. May increase respiration and heart rate. Vapor may displace oxygen and cause suffocation.	N/A	N/A	N/A	N/A
Hydrochloric acid	L	Corrosive to Metals (Category 1) Skin corrosion (Category 1B) Serious eye damage (Category 1)	CaO NaOH Metals KMnO ₄	N/A	N/A	N/A
Sodium hydroxide	S	Corrosive to metals (Category 1) Skin corrosion (Category 1B) Serious eye damage (Category 1)	HCl Aluminum	N/A	N/A	N/A
Potassium Permanganate	S	Oxidizing solids (Category 2) Acute oral toxicity (Category 4) Skin corrosion/Irritation (Category 1C) Serious eye damage/Eye irritation (Category 1) Reproductive Toxicity (Category 2) Respiratory system, Central nervous system (single exposure) (Category 3) Brain toxicity (repeated exposure) (Category 2)	HCl	N/A	N/A	N/A

*Complete this section of the EDP using the Globally Harmonized System (GHS) hazard classifications that are indicated on Section 2 of your chemical's SDS. This is NOT the same hazard ranking system as the National Fire Protection Association (NFPA) "Fire Diamond". Note that the GHS hazard ranking system runs opposite to the NFPA "Fire Diamond" system. For example, a Category 4 NFPA hazard is the highest in severity, while a Category 4 GHS hazard is the lowest in severity. For example, for acetic acid (SDS [here](#)) you would write in: Flammable Liquid (Category 3), Skin Corrosion (Category 1A), and Serious Eye Damage (Category 1).

Chemical Toxicology, Regulation and Disposal: List the same chemicals that appear above, in the same order.

Chemical Name	Toxicology			Waste Classification for Disposal	Personal Protective Equipment Specific to this Chemical
	8-Hour TWA	STEL	Ceiling		
Dry ice	9000 mg/m3	54000 mg/m3	N/A	Avoided or minimized	Eye/face protection Skin and body protection Respiratory protection Hygiene measures
Hydrochloric acid	2 ppm	N/A	2 ppm	Hazardous waste	Eye/face protection Skin and body protection Respiratory protection Hygiene measures
Sodium hydroxide	2 mg/m3	N/A	N/A	Hazardous waste	Eye/face protection Skin and body protection Respiratory protection Hygiene measures
Potassium Permanganate	0.1 mg/m3	3 mg/m3	5 mg/m3	Hazardous waste	Eye/face protection Skin and body protection Respiratory protection Hygiene measures

Biohazards: Provide details below on any biological hazards that may occur during the design, development, preparation or competition phases of your car. Please list the biological hazards, the biohazard level, and a description of how these agents will be safety handled.

N/A

Standard / Safe Operating Procedures Page

- Provide step-by-step details for each of the sections shown below.
- Identify the hazards, the control methods and the personal protective equipment (PPE) required.
- Provide adequate detail so that the reviewers of this document will have adequate understanding of your procedure to pass judgment on the safety of your vehicle.

The **Emergency Shutdown** section should have only one or two steps required to stop your vehicle and bring it to a safe state.

The **Start-Up Procedure** section should list all the steps required to prepare your chemicals and vehicle at your preparation table. This must include procedures for making each solution required for your car to operate.

The **Run Time Procedure** should describe all steps to operate your vehicle at the competition starting line, including activation of any switches, valves, etc.

The **Shutdown Procedure** should describe the steps normally taken to shut down your vehicle at the end of your competitive run. This cannot involve disconnecting tubing, removing components of the car, or cutting electrical connections on the race floor.

The **Cleanup / Waste Disposal** section should list all the steps required to clean your vehicle of all chemicals and proper chemical disposal. Please keep in mind that at the competition, you most likely will not be working in a chemistry laboratory or have access to a sink. You should consider what additional supplies and steps will be needed when you are doing clean up at your table at the competition site. Please also list which waste bucket you will be using to dispose of the waste (example: Acids, Bases, Organic, Aqueous, Inert Solid Waste). This must include steps of how any student made electro chemical (battery) devices will be separated, neutralized and made safe prior to disposal in solid waste streams.

Team must also remember that all glassware must be clearly labeled at the competition site.

Unlabeled chemicals at the tableside is grounds for disqualification.

Sequence of Steps	Potential Hazards	Procedure to Control Hazard	PPE or Equipment Required
Emergency Shutdown			
Stop the vehicle by lifting it up from the ground and disrupt the electrical circuit by removing the banana clamps while wearing rubber gloves while being careful of the hot and cold containers.	If the vehicle is dropped or tilted over, there is a potential spill of the $\text{Ca}(\text{OH})_2$ battery solution as well as the KMnO_4 and NaOH solution.	Make sure to use both hands when carrying the car. Wear nonslip shoes, and watch where you walk.	Rubber Gloves, Insulated Gloves, Lab Coat, Safety Goggles
Start-up Procedure			
Pour the calcium oxide into the hydrochloric acid and sodium hydroxide solution. Inject the sugar-sodium hydroxide solution into the potassium permanganate.	Possible spillage of hot solution onto the color photosensor on	Turn the valve open, making sure there is no KMnO_4 spilling out.	Rubber Gloves, Insulated Gloves, Lab Coat, Safety Goggles

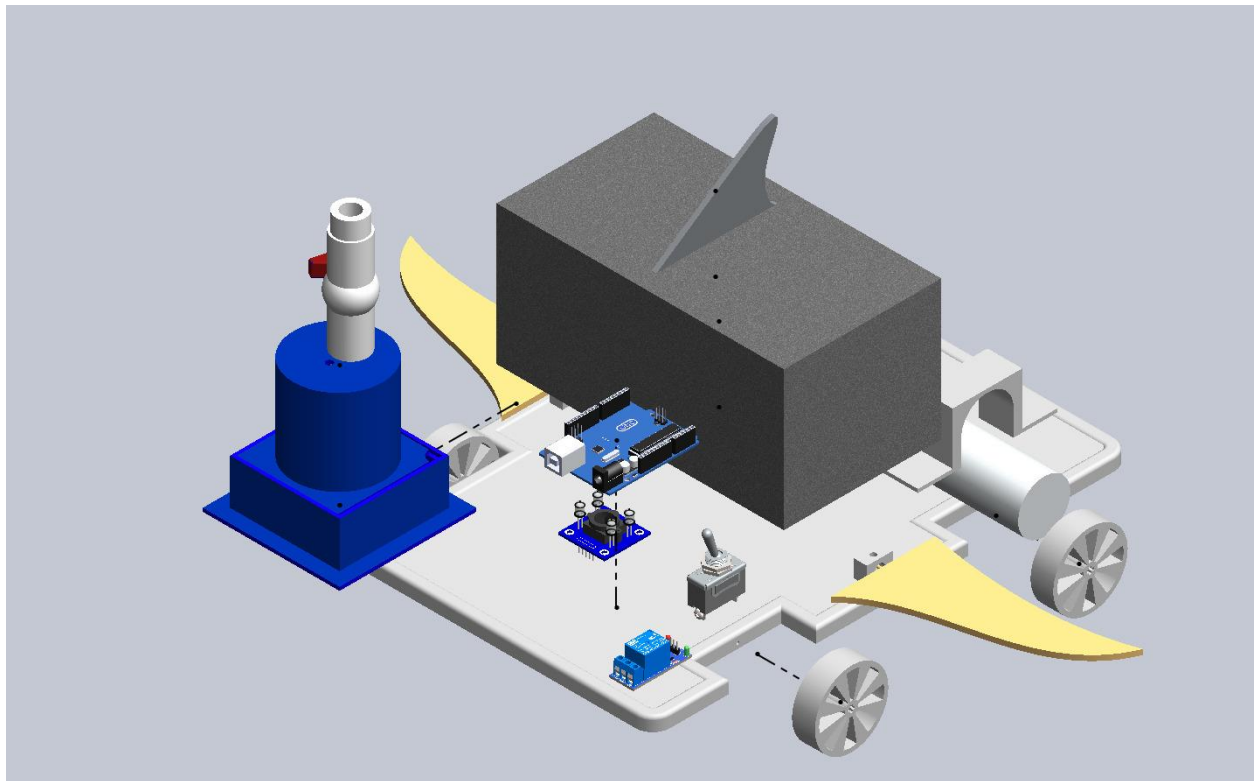
	the Arduino circuit board.		
Run Time Procedure			
Vehicle will run as long as the Chemical chameleon reaction has detected the desired color	Possible spillage of hot solution onto the color photosensor on the Arduino circuit board.	Ensuring the beaker is sufficiently large to reduce the risk of spills. Utilizing 3D printing to ensure a precise fit for the beaker, preventing any potential spillage.	Rubber Gloves, Insulated Gloves, Lab Coat, Safety Goggles
Shutdown Procedure			
The vehicle will cease operation upon detecting a color change through the color sensors. Once the change is registered, the motor will deactivate, halting the car	Possible spillage of hot solution onto the color photosensor on the Arduino circuit board.	Ensuring the beaker is sufficiently large to reduce the risk of spills. Utilizing 3D printing to ensure a precise fit for the beaker, preventing any potential spillage.	Rubber Gloves, Insulated Gloves, Lab Coat, Safety Goggles
Cleanup / Waste Disposal			
The Propulsion solution is caustic waste which will be disposed of into a basic waste container neutralized with a weak acid (acetic acid). The mesh can be reused for future batteries. The Iodine Clock Solution will be stored in a labeled organic waste sealed container. All the containers will be labeled	The reaction is hot. There's a possibility of spillage from containers holding either $\text{Ca}(\text{OH})_2$ or the Chemical Chameleon solution.	Gradually adding diluted hydrochloric acid over time to neutralize the entire $\text{Ca}(\text{OH})_2$ solution. Ensure that containers are tightly sealed.	Rubber Gloves, Insulated Gloves, Lab Coat, Safety Goggles

Equipment List

- Please list every piece of equipment on the car.
- Please include all manufacturer's specification documents or specifications for custom-built components in the EDP Supplement document.

Equipment	Manufacturer	Operating Limits: Temperature	Operating Limits: Pressure	Materials incompatible with chemicals?
EG Module Heatsink Semiconductor	Peltier	N/A	N/A	N/A
Arduino Uno R3 Board	Arduino	N/A	N/A	N/A
PVC Valve	Everbilt	N/A	N/A	N/A
TCS 3200 Color Sensor	Arduino	N/A	N/A	N/A
LED Strip	GXILEE	N/A	N/A	N/A
Aluminum Heatsink	Awxlumv	N/A	N/A	N/A

Note: It is highly recommended that you include a **CAD** or other diagram indicating where each piece of equipment is located within the design of your car. Use the same naming scheme in the diagram as in the equipment list.”



Discharged Hydrogen (or other flammable chemicals) Calculations

If your car will include a small amount of hydrogen discharge, please use this space to provide calculations to prove to the reviewer that any discharged hydrogen is well below the Lower Flammable Limit (LFL)/Lower Explosive Limit (LEL) of hydrogen for the given volume of the reactor, chamber or fuel cell in which hydrogen is stored. *If your car does not use hydrogen, then this section should be left blank.*

Pressure Calculations

For all cars with pressure greater than 5psig, please complete the following in this section. The textbook “Chemical Process Safety” by Crowl and Louvar can be used as reference. Please see Appendix A of the Safety Rules for full instructions on what is required for Pressure Testing. *If your car does not generate pressure above 5 psig, you may leave this section blank.*

- List of potential credible overpressure scenario(s) (i.e., too much reactant added, blocked valve, etc.)
- Sizing calculations for a pressure relief device. You must calculate both the appropriate pressure relief set point and orifice size/device capacity based on the worst-case overpressure scenario identified above;
- Test procedure and results for a pressure relief.

Capital Cost of Vehicle Calculations

Referencing the competition rule surrounding Capital Cost of Vehicle, use the space below to show the capital cost of your vehicle and all related calculations.

Product	Qty	Price	Amount Used	Factor	Price of Amount Used	
Carbon Fiber Filament	1 kg	\$94.99	150 g	0.1500	\$	14.25
PLA Filament	1 kg	\$19.99	746.2 g	0.7462	\$	14.92
PVC Valve	1 pc	\$2.98	3 pc	3.0000	\$	8.94
TEGs	6 pcs	\$23.19	8 pcs	1.2500	\$	28.99
Thermal Paste	1 pc	\$11.99	2 pcs	2.0000	\$	23.98
Heatsinks	4 pcs	\$9.99	8 pcs	2.0000	\$	19.98
HCl	2.5 L	\$107.00	400 mL	0.1600	\$	17.12
NaOH	5 lbs	\$42.87	11 g	0.0049	\$	0.21
Dry Ice	6.9 lb	\$18.67	150 g	2.0000	\$	37.34
Color Sensor	1 pc	\$9.99	1 pc	1.0000	\$	9.99
Arduino	1 pc	\$16.99	1 pc	1.0000	\$	16.99
Relay	1 pc	\$6.23	1 pc	1.0000	\$	6.23
Epoxy	2 pc	\$14.99	1 pcs	1.0000	\$	14.99
Switch	1 pc	\$5.29	1 pc	1.0000	\$	5.29
KMnO4	8 oz	\$15.48	.19 g	0.0008	\$	0.01
More NaOH	1 L	\$22.49	5 ml	0.0050	\$	0.11
Total					\$	219.34

Team Waste Tags

Please fill out a waste tag fully describing each of your competition day waste streams, using the form below. This must be completed and included with your EDP. You must print and bring enough copies to accommodate all the waste you might generate. A completely filled out waste tag will be required before any team is allowed to dispose of waste into an AIChE waste collection system.

ALL SOLID WASTE MUST BE DISASSEMBLED, MADE SAFE AND BE LIQUID FREE

Team Name: Sharkus Electrus	No.: 1
Waste Description: Aqueous Acid-Base solution	
Composition:	
Amount (ml/g)	Description
300 ml + 11 g	Acidic solution of HCl + NaOH

Disposal Waste Stream (select only 1):

☐ Flammable ☐ Organic
☒ Acid ☐ Base
☐ Aqueous (pH 6-9) ☐ Solids (pH 6-9)

Notes: Still warm to the touch

Team Name: Sharkus Electrus	No.: 2
Waste Description: Alkaline solution containing manganese compounds and organic byproducts	
Composition:	
Amount (ml/g)	Description
37 ml	Alkaline solution

Disposal Waste Stream (select only 1):

☐ Flammable ☐ Organic
☐ Acid ☐ Base
☒ Aqueous (pH 6-9) ☐ Solids (pH 6-9)

Notes:



2025 Engineering Design Package (EDP)



2025 Engineering Design Package (EDP)

SAFETY DATA SHEET

Carbon Dioxide, Solid or Dry Ice

Section 1. Identification

GHS product identifier	: Carbon Dioxide, Solid or Dry Ice
Chemical name	: Carbon dioxide, solid
Other means of identification	: Dry ice; carbonic anhydride
Product type	: Solid.
Product use	: Synthetic/Analytical chemistry.
Synonym	: Dry ice; carbonic anhydride
SDS #	: 001091
Supplier's details	: Airgas USA, LLC and its affiliates 259 North Radnor-Chester Road Suite 100 Radnor, PA 19087-5283 1-610-687-5253
24-hour telephone	: 1-866-734-3438

Section 2. Hazards identification

OSHA/HCS status	: Not classified.
Classification of the substance or mixture	: Not classified by Globally Harmonized System of Classification and Labeling (GHS).

GHS label elements

Signal word	: Danger
Hazard statements	: May displace oxygen and cause rapid suffocation. May increase respiration and heart rate. May cause cryogenic burns or injury. May cause frostbite. May increase respiration and heart rate.

Precautionary statements

General	: Read label before use. Keep out of reach of children. If medical advice is needed, have product container or label at hand.
Prevention	: Not applicable.
Response	: Not applicable.
Disposal	: Not applicable.
Hazards not otherwise classified	: Contact with cryogenic liquid can cause frostbite and cryogenic burns. None known.

Section 3. Composition/information on ingredients

Substance/mixture	: Substance
Chemical name	: Carbon dioxide, solid
Other means of identification	: Dry ice; carbonic anhydride
Product code	: 001091

CAS number/other identifiers

CAS number	: 124-38-9
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Section 3. Composition/information on ingredients

Ingredient name	%	CAS number
Carbon Dioxide	100	124-38-9

Any concentration shown as a range is to protect confidentiality or is due to batch variation.

There are no additional ingredients present which, within the current knowledge of the supplier and in the concentrations applicable, are classified as hazardous to health or the environment and hence require reporting in this section.

Occupational exposure limits, if available, are listed in Section 8.

Section 4. First aid measures

Description of necessary first aid measures

- Eye contact** : Immediately flush eyes with plenty of water, occasionally lifting the upper and lower eyelids. Check for and remove any contact lenses. Get medical attention if irritation occurs.
- Inhalation** : Remove victim to fresh air and keep at rest in a position comfortable for breathing. Get medical attention if symptoms occur.
- Skin contact** : Flush contaminated skin with plenty of water. Remove contaminated clothing and shoes. Get medical attention if symptoms occur.
- Ingestion** : Wash out mouth with water. Remove victim to fresh air and keep at rest in a position comfortable for breathing. If material has been swallowed and the exposed person is conscious, give small quantities of water to drink. Do not induce vomiting unless directed to do so by medical personnel. Get medical attention if symptoms occur.

Most important symptoms/effects, acute and delayed

Potential acute health effects

- Eye contact** : May cause eye irritation.
- Inhalation** : May be harmful if inhaled. May cause respiratory irritation.
- Skin contact** : Harmful if absorbed through the skin. May cause skin irritation.
- Frostbite** : Try to warm up the frozen tissues and seek medical attention.
- Ingestion** : May be harmful if swallowed and enters airways.

Over-exposure signs/symptoms

- Eye contact** : No specific data.
- Inhalation** : No specific data.
- Skin contact** : No specific data.
- Ingestion** : No specific data.

Indication of immediate medical attention and special treatment needed, if necessary

- Notes to physician** : Treat symptomatically. Contact poison treatment specialist immediately if large quantities have been ingested or inhaled.
- Specific treatments** : No specific treatment.
- Protection of first-aiders** : No action shall be taken involving any personal risk or without suitable training.

See toxicological information (Section 11)

Section 5. Fire-fighting measures

Extinguishing media

- Suitable extinguishing media** : Use an extinguishing agent suitable for the surrounding fire.
- Unsuitable extinguishing media** : None known.

Section 5. Fire-fighting measures

- Specific hazards arising from the chemical** : No specific fire or explosion hazard.
- Hazardous thermal decomposition products** : Decomposition products may include the following materials:
carbon dioxide
carbon monoxide
- Special protective actions for fire-fighters** : Promptly isolate the scene by removing all persons from the vicinity of the incident if there is a fire. No action shall be taken involving any personal risk or without suitable training.
- Special protective equipment for fire-fighters** : Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode.

Section 6. Accidental release measures

Personal precautions, protective equipment and emergency procedures

- For non-emergency personnel** : No action shall be taken involving any personal risk or without suitable training. Evacuate surrounding areas. Keep unnecessary and unprotected personnel from entering. Do not touch or walk through spilled material. Put on appropriate personal protective equipment.
- For emergency responders** : If specialized clothing is required to deal with the spillage, take note of any information in Section 8 on suitable and unsuitable materials. See also the information in "For non-emergency personnel".
- Environmental precautions** : Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers. Inform the relevant authorities if the product has caused environmental pollution (sewers, waterways, soil or air).

Methods and materials for containment and cleaning up

- Small spill** : Move containers from spill area. Vacuum or sweep up material and place in a designated, labeled waste container. Dispose of via a licensed waste disposal contractor.
- Large spill** : Move containers from spill area. Prevent entry into sewers, water courses, basements or confined areas. Vacuum or sweep up material and place in a designated, labeled waste container. Dispose of via a licensed waste disposal contractor. Note: see Section 1 for emergency contact information and Section 13 for waste disposal.

Section 7. Handling and storage

Precautions for safe handling

- Protective measures** : Put on appropriate personal protective equipment (see Section 8).
- Advice on general occupational hygiene** : Eating, drinking and smoking should be prohibited in areas where this material is handled, stored and processed. Workers should wash hands and face before eating, drinking and smoking. Remove contaminated clothing and protective equipment before entering eating areas. See also Section 8 for additional information on hygiene measures.
- Conditions for safe storage, including any incompatibilities** : Store in accordance with local regulations. Store in original container protected from direct sunlight in a dry, cool and well-ventilated area, away from incompatible materials (see Section 10) and food and drink. Keep container tightly closed and sealed until ready for use. Containers that have been opened must be carefully resealed and kept upright to prevent leakage. Do not store in unlabeled containers. Use appropriate containment to avoid environmental contamination. See Section 10 for incompatible materials before handling or use.

Section 8. Exposure controls/personal protection

Control parameters

Occupational exposure limits

Ingredient name	Exposure limits
Carbon Dioxide	ACGIH TLV (United States, 3/2019). STEL: 54000 mg/m³ 15 minutes. STEL: 30000 ppm 15 minutes. TWA: 9000 mg/m³ 8 hours. TWA: 5000 ppm 8 hours. NIOSH REL (United States, 10/2016). STEL: 54000 mg/m³ 15 minutes. STEL: 30000 ppm 15 minutes. TWA: 9000 mg/m³ 10 hours. TWA: 5000 ppm 10 hours. OSHA PEL (United States, 5/2018). TWA: 9000 mg/m³ 8 hours. TWA: 5000 ppm 8 hours. OSHA PEL 1989 (United States, 3/1989). STEL: 54000 mg/m³ 15 minutes. STEL: 30000 ppm 15 minutes. TWA: 18000 mg/m³ 8 hours. TWA: 10000 ppm 8 hours.

Appropriate engineering controls

: Good general ventilation should be sufficient to control worker exposure to airborne contaminants.

Environmental exposure controls

: Emissions from ventilation or work process equipment should be checked to ensure they comply with the requirements of environmental protection legislation. In some cases, fume scrubbers, filters or engineering modifications to the process equipment will be necessary to reduce emissions to acceptable levels.

Individual protection measures

Hygiene measures

: Wash hands, forearms and face thoroughly after handling chemical products, before eating, smoking and using the lavatory and at the end of the working period. Appropriate techniques should be used to remove potentially contaminated clothing. Wash contaminated clothing before reusing. Ensure that eyewash stations and safety showers are close to the workstation location.

Eye/face protection

: Safety eyewear complying with an approved standard should be used when a risk assessment indicates this is necessary to avoid exposure to liquid splashes, mists, gases or dusts. If contact is possible, the following protection should be worn, unless the assessment indicates a higher degree of protection: safety glasses with side-shields.

Skin protection

Hand protection

: Chemical-resistant, impervious gloves complying with an approved standard should be worn at all times when handling chemical products if a risk assessment indicates this is necessary.

Body protection

: Personal protective equipment for the body should be selected based on the task being performed and the risks involved and should be approved by a specialist before handling this product.

Other skin protection

: Appropriate footwear and any additional skin protection measures should be selected based on the task being performed and the risks involved and should be approved by a specialist before handling this product.

Respiratory protection

: Based on the hazard and potential for exposure, select a respirator that meets the appropriate standard or certification. Respirators must be used according to a respiratory protection program to ensure proper fitting, training, and other important aspects of use.

Section 9. Physical and chemical properties

Appearance

Physical state	: Solid. [WHITE SNOW-LIKE SOLID]
Color	: White.
Odor	: Not available.
Odor threshold	: Not available.
pH	: Not available.
Melting point	: Sublimation temperature: -78.5°C (-109.3 to °F)
Boiling point	: Not available.
Critical temperature	: 31°C (87.8°F)
Flash point	: Not available.
Evaporation rate	: Not available.
Flammability (solid, gas)	: Not available.
Lower and upper explosive (flammable) limits	: Not available.
Vapor pressure	: Not available.
Vapor density	: Not available.
Specific Volume (ft ³ /lb)	: 0.6579
Gas Density (lb/ft ³)	: 1.52
Relative density	: Density Solid (Dry Ice) 97.5189 lb./ft.3 at -109.3° F
Solubility	: Not available.
Solubility in water	: Not available.
Partition coefficient: n-octanol/water	: Not available.
Auto-ignition temperature	: Not available.
Decomposition temperature	: Not available.
Viscosity	: Not available.
Flow time (ISO 2431)	: Not available.
Molecular weight	: 44.01 g/mole

Section 10. Stability and reactivity

Reactivity	: No specific test data related to reactivity available for this product or its ingredients.
Chemical stability	: The product is stable.
Possibility of hazardous reactions	: Under normal conditions of storage and use, hazardous reactions will not occur.
Conditions to avoid	: No specific data.
Incompatible materials	: No specific data.
Hazardous decomposition products	: Under normal conditions of storage and use, hazardous decomposition products should not be produced.
Hazardous polymerization	: Under normal conditions of storage and use, hazardous polymerization will not occur.

Section 11. Toxicological information

Information on toxicological effects

Acute toxicity

Not available.

Irritation/Corrosion

Not available.

Sensitization

Not available.

Mutagenicity

Not available.

Carcinogenicity

Not available.

Reproductive toxicity

Not available.

Teratogenicity

Not available.

Specific target organ toxicity (single exposure)

Not available.

Specific target organ toxicity (repeated exposure)

Not available.

Aspiration hazard

Not available.

Information on the likely routes of exposure : Not available.

Potential acute health effects

Eye contact : May cause eye irritation.
Inhalation : May be harmful if inhaled. May cause respiratory irritation.
Skin contact : Harmful if absorbed through the skin. May cause skin irritation.
Ingestion : May be harmful if swallowed and enters airways.

Symptoms related to the physical, chemical and toxicological characteristics

Eye contact : No specific data.
Inhalation : No specific data.
Skin contact : No specific data.
Ingestion : No specific data.

Delayed and immediate effects and also chronic effects from short and long term exposure

Short term exposure

Potential immediate effects : Not available.
Potential delayed effects : Not available.

Long term exposure

Potential immediate effects : Not available.
Potential delayed effects : Not available.

Potential chronic health effects

Section 11. Toxicological information

Not available.

General	: No known significant effects or critical hazards.
Carcinogenicity	: No known significant effects or critical hazards.
Mutagenicity	: No known significant effects or critical hazards.
Teratogenicity	: No known significant effects or critical hazards.
Developmental effects	: No known significant effects or critical hazards.
Fertility effects	: No known significant effects or critical hazards.

Numerical measures of toxicity

Acute toxicity estimates

Not available.

Section 12. Ecological information

Toxicity

Not available.

Persistence and degradability

Not available.

Bioaccumulative potential

Not available.

Mobility in soil

Soil/water partition coefficient (K_{oc})	: Not available.
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Other adverse effects	: No known significant effects or critical hazards.
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Section 13. Disposal considerations

Disposal methods	: The generation of waste should be avoided or minimized wherever possible. Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements. Dispose of surplus and non-recyclable products via a licensed waste disposal contractor. Waste should not be disposed of untreated to the sewer unless fully compliant with the requirements of all authorities with jurisdiction. Waste packaging should be recycled. Incineration or landfill should only be considered when recycling is not feasible. This material and its container must be disposed of in a safe way. Empty containers or liners may retain some product residues. Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers.
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Section 14. Transport information

Section 14. Transport information

	DOT	TDG	Mexico	IMDG	IATA
UN number	UN1845	UN1845	UN1845	UN1845	UN1845
UN proper shipping name	CARBON DIOXIDE, SOLID OR DRY ICE	CARBON DIOXIDE, SOLID; OR DRY ICE	CARBON DIOXIDE, SOLID OR DRY ICE	CARBON DIOXIDE, SOLID (DRY ICE)	CARBON DIOXIDE, SOLID
Transport hazard class(es)	9 	9 	9 	9 	9 
Packing group	-	-	-	-	-
Environmental hazards	No.	No.	No.	No.	No.

“Refer to CFR 49 (or authority having jurisdiction) to determine the information required for shipment of the product.”

Additional information

DOT Classification

: **Limited quantity** Yes.

Quantity limitation Passenger aircraft/rail: 200 kg. Cargo aircraft: 200 kg.

TDG Classification

: Product classified as per the following sections of the Transportation of Dangerous Goods Regulations: 2.43-2.45 (Class 9).

Explosive Limit and Limited Quantity Index 5

Passenger Carrying Vessel Index 200

Special provisions 18

IATA

: **Quantity limitation** Passenger and Cargo Aircraft: 200 kg. Cargo Aircraft Only: 200 kg.

Special precautions for user

: **Transport within user's premises:** always transport in closed containers that are upright and secure. Ensure that persons transporting the product know what to do in the event of an accident or spillage.

Transport in bulk according to IMO instruments

: Not available.

Section 15. Regulatory information

U.S. Federal regulations

: **TSCA 8(a) CDR Exempt/Partial exemption:** This material is listed or exempted.

Clean Air Act Section 112 (b) Hazardous Air Pollutants (HAPs)

: Not listed

Clean Air Act Section 602 Class I Substances

: Not listed

Clean Air Act Section 602 Class II Substances

: Not listed

DEA List I Chemicals (Precursor Chemicals)

: Not listed

DEA List II Chemicals (Essential Chemicals)

: Not listed

SARA 302/304

Composition/information on ingredients

No products were found.

Section 15. Regulatory information

SARA 304 RQ : Not applicable.

SARA 311/312

Classification : Refer to Section 2: Hazards Identification of this SDS for classification of substance.

State regulations

Massachusetts : This material is listed.

New York : This material is not listed.

New Jersey : This material is listed.

Pennsylvania : This material is listed.

California Prop. 65

This product does not require a Safe Harbor warning under California Prop. 65.

International regulations

Chemical Weapon Convention List Schedules I, II & III Chemicals

Not listed.

Montreal Protocol

Not listed.

Stockholm Convention on Persistent Organic Pollutants

Not listed.

Rotterdam Convention on Prior Informed Consent (PIC)

Not listed.

UNECE Aarhus Protocol on POPs and Heavy Metals

Not listed.

Inventory list

Australia : This material is listed or exempted.

Canada : This material is listed or exempted.

China : This material is listed or exempted.

Europe : This material is listed or exempted.

Japan : **Japan inventory (ENCS)**: This material is listed or exempted.
Japan inventory (ISHL): This material is listed or exempted.

New Zealand : This material is listed or exempted.

Philippines : This material is listed or exempted.

Republic of Korea : This material is listed or exempted.

Taiwan : This material is listed or exempted.

Thailand : Not determined.

Turkey : This material is listed or exempted.

United States : This material is active or exempted.

Viet Nam : This material is listed or exempted.

Section 16. Other information

Hazardous Material Information System (U.S.A.)

Health	/	3
Flammability		0
Physical hazards		0

Section 16. Other information

Caution: HMIS® ratings are based on a 0-4 rating scale, with 0 representing minimal hazards or risks, and 4 representing significant hazards or risks. Although HMIS® ratings and the associated label are not required on SDSs or products leaving a facility under 29 CFR 1910.1200, the preparer may choose to provide them. HMIS® ratings are to be used with a fully implemented HMIS® program. HMIS® is a registered trademark and service mark of the American Coatings Association, Inc.

The customer is responsible for determining the PPE code for this material. For more information on HMIS® Personal Protective Equipment (PPE) codes, consult the HMIS® Implementation Manual.

National Fire Protection Association (U.S.A.)



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Copyright ©2001, National Fire Protection Association, Quincy, MA 02269. This warning system is intended to be interpreted and applied only by properly trained individuals to identify fire, health and reactivity hazards of chemicals. The user is referred to certain limited number of chemicals with recommended classifications in NFPA 49 and NFPA 325, which would be used as a guideline only. Whether the chemicals are classified by NFPA or not, anyone using the 704 systems to classify chemicals does so at their own risk.

Procedure used to derive the classification

Classification	Justification
Not classified.	

History

Date of printing : 1/27/2022

Date of issue/Date of revision : 1/27/2022

Date of previous issue : 1/27/2022

Version : 1.03

Key to abbreviations : ATE = Acute Toxicity Estimate
BCF = Bioconcentration Factor
GHS = Globally Harmonized System of Classification and Labelling of Chemicals
IATA = International Air Transport Association
IBC = Intermediate Bulk Container
IMDG = International Maritime Dangerous Goods
LogPow = logarithm of the octanol/water partition coefficient
MARPOL = International Convention for the Prevention of Pollution From Ships, 1973 as modified by the Protocol of 1978. ("Marpol" = marine pollution)
UN = United Nations

References : Not available.

Notice to reader

To the best of our knowledge, the information contained herein is accurate. However, neither the above-named supplier, nor any of its subsidiaries, assumes any liability whatsoever for the accuracy or completeness of the information contained herein.

Final determination of suitability of any material is the sole responsibility of the user. All materials may present unknown hazards and should be used with caution. Although certain hazards are described herein, we cannot guarantee that these are the only hazards that exist.

SAFETY DATA SHEET

Creation Date 24-Aug-2009

Revision Date 13-Oct-2023

Revision Number 8

1. Identification

Product Name Hydrochloric acid

Cat No. : A481-212; A481-212LC; S71942SC; S71943; S71943ND; S80036; S80038; SA49

CAS No 7647-01-0
Synonyms Muriatic acid; Hydrogen chloride; HCl (Technical/Certified ACS Plus/Optima/NF/FCC)

Recommended Use Laboratory chemicals.
Uses advised against Food, drug, pesticide or biocidal product use.

Details of the supplier of the safety data sheet

Company

Fisher Scientific Company
One Reagent Lane
Fair Lawn, NJ 07410
Tel: (201) 796-7100

Emergency Telephone Number

CHEMTREC®, Inside the USA: 800-424-9300
CHEMTREC®, Outside the USA: 001-703-527-3887

2. Hazard(s) identification

Classification

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

Corrosive to metals	Category 1
Skin Corrosion/Irritation	Category 1 B
Serious Eye Damage/Eye Irritation	Category 1
Specific target organ toxicity (single exposure)	Category 3
Target Organs - Respiratory system.	

Label Elements

Signal Word

Danger

Hazard Statements

May be corrosive to metals

Causes severe skin burns and eye damage
May cause respiratory irritation

**Precautionary Statements****Prevention**

Do not breathe dust/fume/gas/mist/vapors/spray
Wash face, hands and any exposed skin thoroughly after handling
Wear protective gloves/protective clothing/eye protection/face protection
Use only outdoors or in a well-ventilated area
Keep only in original container

Response

Immediately call a POISON CENTER or doctor/physician

Inhalation

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing

Skin

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower
Wash contaminated clothing before reuse

Eyes

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

Ingestion

IF SWALLOWED: Rinse mouth. DO NOT induce vomiting

Spills

Absorb spillage to prevent material damage

Storage

Store locked up
Store in a well-ventilated place. Keep container tightly closed
Store in corrosive resistant polypropylene container with a resistant inliner
Store in a dry place

Disposal

Dispose of contents/container to an approved waste disposal plant

Hazards not otherwise classified (HNOC)

None identified

3. Composition/Information on Ingredients

Component	CAS No	Weight %
Water	7732-18-5	62-65
Hydrochloric acid	7647-01-0	35-38

4. First-aid measures

Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.
Immediate medical attention is required.

Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

Inhalation

Remove to fresh air. If breathing is difficult, give oxygen. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a

pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.

Ingestion

Do NOT induce vomiting. Call a physician or poison control center immediately.

Most important symptoms and effects

Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

Notes to Physician

Treat symptomatically

5. Fire-fighting measures

Suitable Extinguishing Media

Substance is nonflammable; use agent most appropriate to extinguish surrounding fire.

Unsuitable Extinguishing Media

No information available

Flash Point

No information available

Method -

No information available

Autoignition Temperature

No information available

Explosion Limits**Upper**

No data available

Lower

No data available

Sensitivity to Mechanical Impact

No information available

Sensitivity to Static Discharge

No information available

Specific Hazards Arising from the Chemical

Corrosive material. Causes burns by all exposure routes. Thermal decomposition can lead to release of irritating gases and vapors.

Hazardous Combustion Products

Hydrogen chloride gas.

Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

NFPA

Health
3

Flammability
0

Instability
0

Physical hazards
N/A

6. Accidental release measures

Personal Precautions

Use personal protective equipment as required. Ensure adequate ventilation. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Do not get in eyes, on skin, or on clothing.

Environmental Precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

Methods for Containment and Clean Up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

7. Handling and storage

Handling

Wear personal protective equipment/face protection. Do not breathe mist/vapors/spray. Do not get in eyes, on skin, or on clothing. Do not ingest. If swallowed then seek immediate medical assistance.

Storage.

Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area. Incompatible Materials. Metals. Strong oxidizing agents. Bases. sodium hypochlorite. Amines. Fluorine. Cyanides. Alkaline.

8. Exposure controls / personal protection

Exposure Guidelines

Component	ACGIH TLV	OSHA PEL	NIOSH	Mexico OEL (TWA)
Hydrochloric acid	Ceiling: 2 ppm	Ceiling: 5 ppm Ceiling: 7 mg/m ³ (Vacated) Ceiling: 5 ppm (Vacated) Ceiling: 7 mg/m ³	IDLH: 50 ppm Ceiling: 5 ppm Ceiling: 7 mg/m ³	Ceiling: 2 ppm

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location.

Personal Protective Equipment

Eye/face Protection

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure.

Respiratory Protection

Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

Recommended Filter type:

Particulates filter conforming to EN 143. or. Acid gases filter: Type E, Yellow.

Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

9. Physical and chemical properties

Physical State	Liquid
Appearance	Colorless
Odor	pungent
Odor Threshold	No information available
pH	< 1
Melting Point/Range	-35 °C / -31 °F
Boiling Point/Range	57 °C / 135 °F @ 760 mmHg
Flash Point	No information available
Evaporation Rate	No information available
Flammability (solid,gas)	Not applicable
Flammability or explosive limits	
Upper	No data available
Lower	No data available
Vapor Pressure	125 mbar @ 20 °C
Vapor Density	1.27
Specific Gravity	1.18
Solubility	Soluble in water
Partition coefficient; n-octanol/water	No data available
Autoignition Temperature	No information available
Decomposition Temperature	No information available
Viscosity	1.8 mPa.s @ 15°C
Molecular Formula	HCl

Molecular Weight 55.55

10. Stability and reactivity

Reactive Hazard	None known, based on information available
Stability	Stable under normal conditions.
Conditions to Avoid	Incompatible products. Excess heat.
Incompatible Materials	Metals, Strong oxidizing agents, Bases, sodium hypochlorite, Amines, Fluorine, Cyanides, Alkaline
Hazardous Decomposition Products	Hydrogen chloride gas
Hazardous Polymerization	Hazardous polymerization does not occur.
Hazardous Reactions	Contact with metals may evolve flammable hydrogen gas.

11. Toxicological information

Acute Toxicity

Product Information

Oral LD50

Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

Dermal LD50

Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

Vapor LC50

Based on ATE data, the classification criteria are not met. ATE > 20 mg/l.

Component Information

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Water	-	-	-
Hydrochloric acid	238 - 277 mg/kg (Rat)	> 5010 mg/kg (Rabbit)	1.68 mg/L (Rat) 1 h

Toxicologically Synergistic Products No information available

Delayed and immediate effects as well as chronic effects from short and long-term exposure

Irritation Causes burns by all exposure routes

Sensitization No information available

Carcinogenicity The table below indicates whether each agency has listed any ingredient as a carcinogen.

Component	CAS No	IARC	NTP	ACGIH	OSHA	Mexico
Water	7732-18-5	Not listed	Not listed	Not listed	Not listed	Not listed
Hydrochloric acid	7647-01-0	Not listed	Not listed	Not listed	Not listed	Not listed

IARC (International Agency for Research on Cancer)

IARC (International Agency for Research on Cancer)

Group 1 - Carcinogenic to Humans

Group 2A - Probably Carcinogenic to Humans

Group 2B - Possibly Carcinogenic to Humans

Mutagenic Effects No information available

Reproductive Effects No information available.

Developmental Effects No information available.

Teratogenicity No information available.

STOT - single exposure Respiratory system

STOT - repeated exposure None known

Aspiration hazard No information available

Symptoms / effects, both acute and delayed Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

Endocrine Disruptor Information No information available

Other Adverse Effects The toxicological properties have not been fully investigated.

12. Ecological information

Ecotoxicity

Do not empty into drains. Large amounts will affect pH and harm aquatic organisms.

Component	Freshwater Algae	Freshwater Fish	Microtox	Water Flea
Hydrochloric acid	-	282 mg/L LC50 96 h Gambusia affinis mg/L LC50 48 h Leuciscus idus	-	56mg/L EC50 72h Daphnia

Persistence and Degradability Persistence is unlikely based on information available.

Bioaccumulation/ Accumulation No information available.

Mobility Will likely be mobile in the environment due to its water solubility.

13. Disposal considerations

Waste Disposal Methods Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

14. Transport information

DOT

UN-No UN1789
 Proper Shipping Name HYDROCHLORIC ACID
 Hazard Class 8
 Packing Group II

TDG

UN-No UN1789
 Proper Shipping Name HYDROCHLORIC ACID
 Hazard Class 8
 Packing Group II

IATA

UN-No UN1789
 Proper Shipping Name Hydrochloric acid
 Hazard Class 8
 Packing Group II

IMDG/IMO

UN-No UN1789
 Proper Shipping Name Hydrochloric acid
 Hazard Class 8
 Packing Group II

15. Regulatory information

United States of America Inventory

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	TSCA - EPA Regulatory Flags
Water	7732-18-5	X	ACTIVE	-
Hydrochloric acid	7647-01-0	X	ACTIVE	-

Legend:

TSCA US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

- - Not Listed

TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)

Not applicable

TSCA 12(b) - Notices of Export

Not applicable

International Inventories

Canada (DSL/NDL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

Component	CAS No	DSL	NDL	EINECS	PICCS	ENCS	ISHL	AICS	IECSC	KECL
Water	7732-18-5	X	-	231-791-2	X	X		X	X	KE-35400
Hydrochloric acid	7647-01-0	X	-	231-595-7	X	X	X	X	X	KE-20189

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)**U.S. Federal Regulations****SARA 313**

Component	CAS No	Weight %	SARA 313 - Threshold Values %
Hydrochloric acid	7647-01-0	35-38	1.0

SARA 311/312 Hazard Categories See section 2 for more information**CWA (Clean Water Act)**

Component	CWA - Hazardous Substances	CWA - Reportable Quantities	CWA - Toxic Pollutants	CWA - Priority Pollutants
Hydrochloric acid	X	5000 lb	-	-

Clean Air Act

Component	HAPS Data	Class 1 Ozone Depletors	Class 2 Ozone Depletors
Hydrochloric acid	X		-

OSHA - Occupational Safety and Health Administration

Not applicable

Component	Specifically Regulated Chemicals	Highly Hazardous Chemicals
Hydrochloric acid	-	TQ: 5000 lb

CERCLA

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302)

Component	Hazardous Substances RQs	CERCLA EHS RQs
Hydrochloric acid	5000 lb	5000 lb

California Proposition 65

This product does not contain any Proposition 65 chemicals.

U.S. State Right-to-Know Regulations

Component	Massachusetts	New Jersey	Pennsylvania	Illinois	Rhode Island
Water	-	-	X	-	-
Hydrochloric acid	X	X	X	X	X

U.S. Department of Transportation

Reportable Quantity (RQ): Y
 DOT Marine Pollutant N
 DOT Severe Marine Pollutant N

U.S. Department of Homeland Security

This product contains the following DHS chemicals:

Legend - STQs = Screening Threshold Quantities, APA = A placarded amount

Component	DHS Chemical Facility Anti-Terrorism Standard
Hydrochloric acid	Release STQs - 15000lb (concentration $\geq 37\%$) Release STQs - 5000lb (anhydrous) Theft STQs - 500lb (anhydrous)

Other International Regulations

Mexico - Grade No information available

Authorisation/Restrictions according to EU REACH

Component	CAS No	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
Water	7732-18-5	-	-	-
Hydrochloric acid	7647-01-0	-	Use restricted. See item 75. (see link for restriction details)	-

REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

Safety, health and environmental regulations/legislation specific for the substance or mixture

Component	CAS No	OECD HPV	Persistent Organic Pollutant	Ozone Depletion Potential	Restriction of Hazardous Substances (RoHS)
Water	7732-18-5	Listed	Not applicable	Not applicable	Not applicable
Hydrochloric acid	7647-01-0	Listed	Not applicable	Not applicable	Not applicable

Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Other International Regulations

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements	Rotterdam Convention (PIC)	Basel Convention (Hazardous Waste)
Water	7732-18-5	Not applicable	Not applicable	Not applicable	Not applicable
Hydrochloric acid	7647-01-0	25 tonne	250 tonne	Not applicable	Annex I - Y34

16. Other information

Prepared By

Regulatory Affairs

Thermo Fisher Scientific
Email: EMSDS.RA@thermofisher.com

Creation Date	24-Aug-2009
Revision Date	13-Oct-2023
Print Date	13-Oct-2023
Revision Summary	SDS sections updated. 2. 3. 11.

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

End of SDS

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Sodium Hydroxide, 3M

SECTION 1 : Identification of the substance/mixture and of the supplier

Product name : Sodium Hydroxide, 3M

Manufacturer/Supplier Trade name:

Manufacturer/Supplier Article number: S25884

Recommended uses of the product and uses restrictions on use:

Manufacturer Details:

AquaPhoenix Scientific
9 Barnhart Drive, Hanover, PA 17331

Supplier Details:

Fisher Science Education
15 Jet View Drive, Rochester, NY 14624

Emergency telephone number:

Fisher Science Education Emergency Telephone No.: 800-535-5053

SECTION 2 : Hazards identification

Classification of the substance or mixture:



Corrosive

Corrosive to metals, category 1
Serious eye damage, category 1
Skin corrosion, category 1A

Eye corr. 1
Skin Corr. 1A
Metal Corr. 1

Signal word : Danger

Hazard statements:

May be corrosive to metals
Causes severe skin burns and eye damage
Causes serious eye damage

Precautionary statements:

If medical advice is needed, have product container or label at hand
Keep out of reach of children
Read label before use
Keep only in original container
Wash ... thoroughly after handling
Wear protective gloves/protective clothing/eye protection/face protection
Do not breathe dust/fume/gas/mist/vapours/spray
IF SWALLOWED: Rinse mouth. Do NOT induce vomiting
IF ON SKIN (or hair): Remove/Take off immediately all contaminated clothing. Rinse skin with water/shower
Wash contaminated clothing before reuse
IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing
IF INHALED: Call a POISON CENTER or doctor/physician if you feel unwell
Immediately call a POISON CENTER or doctor/physician
Specific treatment (see supplemental first aid instructions on this label)

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Sodium Hydroxide, 3M

Absorb spillage to prevent material damage
Store in a corrosive resistant/... container with a resistant inner liner
Store locked up
Dispose of contents/container to ...

Other Non-GHS Classification:

WHMIS



NFPA/HMIS



NFPA SCALE (0-4)

Health	2
Flammability	0
Physical Hazard	0
Personal Protection	X

HMIS RATINGS (0-4)

SECTION 3 : Composition/information on ingredients

Ingredients:		
CAS 1310-73-2	Sodium Hydroxide	12 %
CAS 7732-18-5	Deionized Water	88 %
Percentages are by weight		

SECTION 4 : First aid measures

Description of first aid measures

After inhalation: Move exposed individual to fresh air. Loosen clothing as necessary and position individual in a comfortable position. Seek medical advice if discomfort or irritation persists. If breathing difficult, give oxygen.

After skin contact: Take off contaminated clothing and shoes immediately. Wash affected area with soap and water. Seek medical attention if irritation, discomfort persists.

After eye contact: Protect unexposed eye. Rinse/flush exposed eye(s) gently using water for 15-20 minutes. Remove contact lens(es) if able to do so during rinsing. Immediately get medical assistance.

After swallowing: Rinse mouth thoroughly. Do not induce vomiting. Have exposed individual drink sips of water. Seek medical attention if irritation, discomfort or vomiting persists.

Most important symptoms and effects, both acute and delayed:

Irritation, Nausea, Headache, Shortness of breath.;

Indication of any immediate medical attention and special treatment needed:

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Sodium Hydroxide, 3M

If seeking medical attention, provide SDS document to physician.

SECTION 5 : Firefighting measures

Extinguishing media

Suitable extinguishing agents: If in laboratory setting, follow laboratory fire suppression procedures. Use appropriate fire suppression agents for adjacent combustible materials or sources of ignition

For safety reasons unsuitable extinguishing agents: Carbon dioxide. Carbon dioxide.

Special hazards arising from the substance or mixture:

Combustion products may include carbon oxides or other toxic vapors. Thermal decomposition can lead to release of irritating gases and vapors. Sodium oxides.

Advice for firefighters:

Protective equipment: Use NIOSH-approved respiratory protection/breathing apparatus.

Additional information (precautions): Move product containers away from fire or keep cool with water spray as a protective measure, where feasible.

SECTION 6 : Accidental release measures

Personal precautions, protective equipment and emergency procedures:

Wear protective equipment. Transfer to a disposal or recovery container. Use respiratory protective device against the effects of fumes/dust/aerosol. Keep unprotected persons away. Ensure adequate ventilation. Keep away from ignition sources. Protect from heat.

Environmental precautions:

Prevent from reaching drains, sewer or waterway. Collect contaminated soil for characterization per Section 13

Methods and material for containment and cleaning up:

If in a laboratory setting, follow Chemical Hygiene Plan procedures. Place into properly labeled containers for recovery or disposal. If necessary, use trained response staff/contractor. Collect liquid and dilute with water. Neutralize with dilute acid solutions. Decant water to drain with excess water. Absorb with suitable material. Dispose of remaining solid as normal refuse. Always obey local regulations.

Reference to other sections:

SECTION 7 : Handling and storage

Precautions for safe handling:

Absorb spillage to prevent material damage due to corrosiveness to metal. Avoid contact with eyes, skin, and clothing. Wash hands after handling. Do not mix with acids. Follow good hygiene procedures when handling chemical materials. Use only in well ventilated areas.

Conditions for safe storage, including any incompatibilities:

Provide ventilation for containers. Avoid storage near extreme heat, ignition sources or open flame. Store away from foodstuffs. Store away from oxidizing agents. Store in cool, dry conditions in well sealed containers. Store with Corrosives.

SECTION 8 : Exposure controls/personal protection



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Sodium Hydroxide, 3M

Control Parameters:	1310-73-2, Sodium Hydroxide, OSHA PEL TWA 2 mg/m3 1310-73-2, Sodium Hydroxide, ACGIH TLV TWA 2 mg/m3
Appropriate Engineering controls:	Emergency eye wash fountains and safety showers should be available in the immediate vicinity of use/handling. Provide exhaust ventilation or other engineering controls to keep the airborne concentrations of vapor or dusts (total/respirable) below the applicable workplace exposure limits (Occupational Exposure Limits-OELs) indicated above. Use under a chemical fume hood.
Respiratory protection:	Use suitable respiratory protective device when high concentrations are present. Use suitable respiratory protective device when aerosol or mist is formed. For spills, respiratory protection may be advisable. Use under a chemical fume hood.
Protection of skin:	The glove material has to be impermeable and resistant to the product/ the substance/ the preparation being used/handled. Selection of the glove material on consideration of the penetration times, rates of diffusion and the degradation.
Eye protection:	Safety glasses with side shields or goggles.
General hygienic measures:	The usual precautionary measures are to be adhered to when handling chemicals. Keep away from food, beverages and feed sources. Immediately remove all soiled and contaminated clothing. Wash hands before breaks and at the end of work. Do not inhale gases/fumes/dust/mist/vapor/aerosols. Avoid contact with the eyes and skin.

SECTION 9 : Physical and chemical properties

Appearance (physical state,color):	Clear, colorless liquid	Explosion limit lower: Explosion limit upper:	Non Explosive Non Explosive
Odor:	Odorless	Vapor pressure:	14mmHg @ 20C
Odor threshold:	Not Determined	Vapor density:	>1
pH-value:	13.3	Relative density:	Approx 1
Melting/Freezing point:	Approx 0°C	Solubilities:	Soluble in Water
Boiling point/Boiling range:	Approx 100°C	Partition coefficient (n-octanol/water):	Not Determined
Flash point (closed cup):	Not Determined	Auto/Self-ignition temperature:	Not Determined
Evaporation rate:	Not Determined	Decomposition temperature:	Not Determined
Flammability (solid,gaseous):	Not Determined	Viscosity:	a. Kinematic: Not Determined b. Dynamic: Not Determined
Density: Not Determined			

SECTION 10 : Stability and reactivity

Reactivity: solution attacks metals such as aluminium, tin, lead and zinc Also generates heat on exposure to acids. Aqueous solutions react violently with acids.

Chemical stability: No decomposition if used and stored according to specifications.

Possible hazardous reactions:

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Conditions to avoid: Incompatible materials, excess heat

Incompatible materials: acids, Organic materials, Chlorinated solvents, Aluminum, Phosphorus, Tin/tin oxides, Zinc

Hazardous decomposition products: sodium oxides, hydrogen

SECTION 11 : Toxicological information

Acute Toxicity: No additional information.

Chronic Toxicity: No additional information.

Corrosion Irritation:

Ocular:	1310-73-2	Rabbit: Corrosive to eyes
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Dermal:	1310-73-2	Rabbit: Causes Burns
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Sensitization:	No additional information.
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Single Target Organ (STOT):	No additional information.
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Numerical Measures:	No additional information.
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Carcinogenicity:	Not listed as a carcinogen: 1310-73-2
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Mutagenicity:	No additional information.
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Reproductive Toxicity:	No additional information.
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SECTION 12 : Ecological information

Ecotoxicity Persistence and degradability: Readily degradable in the environment.

Bioaccumulative potential: Not Bioaccumulative.

Mobility in soil: -1.87 (water)

Other adverse effects:

SECTION 13 : Disposal considerations

Waste disposal recommendations:

Product/containers must not be disposed together with household garbage. Do not allow product to reach sewage system or open water. It is the responsibility of the waste generator to properly characterize all waste materials according to applicable regulatory entities (US 40CFR262.11). Consult federal state/ provincial and local regulations regarding the proper disposal of waste material that may incorporate some amount of this product. Neutralize with dilute acid solutions.

SECTION 14 : Transport information

UN-Number

1824

UN proper shipping name

Sodium hydroxide solution

Transport hazard class(es)



Class:

8 Corrosive substances

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Sodium Hydroxide, 3M

Packing group:II

Environmental hazard:

Transport in bulk:

Special precautions for user:

SECTION 15 : Regulatory information

United States (USA)

SARA Section 311/312 (Specific toxic chemical listings):

None of the ingredients is listed

SARA Section 313 (Specific toxic chemical listings):

None of the ingredients is listed

RCRA (hazardous waste code):

None of the ingredients is listed

TSCA (Toxic Substances Control Act):

All ingredients are listed.

CERCLA (Comprehensive Environmental Response, Compensation, and Liability Act):

1310-73-2 Sodium Hydroxide 1000 lb

Proposition 65 (California):

Chemicals known to cause cancer:

None of the ingredients is listed

Chemicals known to cause reproductive toxicity for females:

None of the ingredients is listed

Chemicals known to cause reproductive toxicity for males:

None of the ingredients is listed

Chemicals known to cause developmental toxicity:

None of the ingredients is listed

Canada

Canadian Domestic Substances List (DSL):

All ingredients are listed.

Canadian NPRI Ingredient Disclosure list (limit 0.1%):

None of the ingredients is listed

Canadian NPRI Ingredient Disclosure list (limit 1%):

1310-73-2 Sodium Hydroxide

SECTION 16 : Other information

This product has been classified in accordance with hazard criteria of the Controlled Products Regulations and the SDS contains all the information required by the Controlled Products Regulations. Note: The responsibility to provide a safe workplace remains with the user. The user should consider the health hazards and safety information contained herein as a guide and should take those precautions required in an individual operation to instruct employees and develop work practice procedures for a safe work environment. The information contained herein is, to the best of our knowledge and belief, accurate. However, since the conditions of handling and use are beyond our control, we make no guarantee of results, and assume no liability for damages incurred by the use of this

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material. It is the responsibility of the user to comply with all applicable laws and regulations applicable to this material.

GHS Full Text Phrases:

Abbreviations and acronyms:

IMDG: International Maritime Code for Dangerous Goods

PNEC: Predicted No-Effect Concentration (REACH)

CFR: Code of Federal Regulations (USA)

SARA: Superfund Amendments and Reauthorization Act (USA)

RCRA: Resource Conservation and Recovery Act (USA)

TSCA: Toxic Substances Control Act (USA)

NPRI: National Pollutant Release Inventory (Canada)

DOT: US Department of Transportation

IATA: International Air Transport Association

GHS: Globally Harmonized System of Classification and Labelling of Chemicals

ACGIH: American Conference of Governmental Industrial Hygienists

CAS: Chemical Abstracts Service (division of the American Chemical Society)

NFPA: National Fire Protection Association (USA)

HMIS: Hazardous Materials Identification System (USA)

WHMIS: Workplace Hazardous Materials Information System (Canada)

DNEL: Derived No-Effect Level (REACH)

Effective date : 12.14.2014

Last updated : 03.19.2015

SAFETY DATA SHEET

Creation Date 23-Oct-2009

Revision Date 24-Dec-2021

Revision Number 11

1. Identification

Product Name Potassium permanganate

Cat No. : P279-212; P279-500; P287-212; P287-500; NC0553994

CAS No 7722-64-7

Synonyms No information available

Recommended Use Laboratory chemicals.

Uses advised against Food, drug, pesticide or biocidal product use.

Details of the supplier of the safety data sheet

Company

Fisher Scientific Company
One Reagent Lane
Fair Lawn, NJ 07410
Tel: (201) 796-7100

Emergency Telephone Number CHEMTREC®, Inside the USA: 800-424-9300
CHEMTREC®, Outside the USA: 001-703-527-3887

2. Hazard(s) identification

Classification

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

Oxidizing solids	Category 2
Acute oral toxicity	Category 4
Skin Corrosion/Irritation	Category 1 C
Serious Eye Damage/Eye Irritation	Category 1
Reproductive Toxicity	Category 2
Specific target organ toxicity (single exposure)	Category 3
Target Organs - Respiratory system, Central nervous system (CNS).	
Specific target organ toxicity - (repeated exposure)	Category 2
Target Organs - Brain.	

Label Elements

Signal Word
Danger

Hazard Statements

May intensify fire; oxidizer
Harmful if swallowed
Causes severe skin burns and eye damage
May cause respiratory irritation
Suspected of damaging the unborn child
May cause damage to organs through prolonged or repeated exposure

**Precautionary Statements****Prevention**

Obtain special instructions before use
Do not handle until all safety precautions have been read and understood
Use personal protective equipment as required
Wash face, hands and any exposed skin thoroughly after handling
Do not eat, drink or smoke when using this product
Do not breathe dust/fume/gas/mist/vapors/spray
Use only outdoors or in a well-ventilated area
Keep away from heat/sparks/open flames/hot surfaces. - No smoking
Keep/Store away from clothing/ other combustible materials
Take any precaution to avoid mixing with combustibles

Response

Immediately call a POISON CENTER or doctor/physician

Inhalation

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing

Skin

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower
Wash contaminated clothing before reuse

Eyes

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

Ingestion

Rinse mouth

Do NOT induce vomiting

Fire

In case of fire: Use CO₂, dry chemical, or foam for extinction

Storage

Store locked up

Store in a well-ventilated place. Keep container tightly closed

Disposal

Dispose of contents/container to an approved waste disposal plant

Hazards not otherwise classified (HNOC)

Very toxic to aquatic life with long lasting effects

3. Composition/Information on Ingredients

Component	CAS No	Weight %
Potassium permanganate	7722-64-7	>95

4. First-aid measures

General Advice	Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.
Eye Contact	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required. Keep eye wide open while rinsing.
Skin Contact	Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Call a physician immediately.
Inhalation	Remove to fresh air. If not breathing, give artificial respiration. Call a physician or poison control center immediately. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device.
Ingestion	Immediate medical attention is required. Do NOT induce vomiting. Drink plenty of water. Never give anything by mouth to an unconscious person.
Most important symptoms and effects	Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation
Notes to Physician	Treat symptomatically

5. Fire-fighting measures

Suitable Extinguishing Media	CO ₂ , dry chemical, dry sand, alcohol-resistant foam.
Unsuitable Extinguishing Media	No information available
Flash Point	No information available
Method -	No information available
Autoignition Temperature	Not applicable
Explosion Limits	
Upper	No data available
Lower	No data available
Oxidizing Properties	Oxidizer
Sensitivity to Mechanical Impact	No information available
Sensitivity to Static Discharge	No information available

Specific Hazards Arising from the Chemical

The product causes burns of eyes, skin and mucous membranes. Oxidizer: Contact with combustible/organic material may cause fire. May ignite combustibles (wood paper, oil, clothing, etc.). Do not allow run-off from fire-fighting to enter drains or water courses.

Hazardous Combustion Products

Heavy metal oxides. Potassium oxides.

Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

NFPA

Health
3

Flammability
0

Instability
2

Physical hazards
OX

6. Accidental release measures

Personal Precautions	Use personal protective equipment as required. Evacuate personnel to safe areas. Avoid contact with skin, eyes or clothing.
-----------------------------	---

Environmental Precautions Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained. Should not be released into the environment.

Methods for Containment and Clean Up Sweep up and shovel into suitable containers for disposal. Avoid dust formation. Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Sweep up and shovel into suitable containers for disposal.

7. Handling and storage

Handling Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe dust. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from clothing and other combustible materials.

Storage. Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area. Do not store near combustible materials. Incompatible Materials. Reducing Agent. Strong acids. Strong reducing agents. Combustible material.

8. Exposure controls / personal protection

Exposure Guidelines

Component	ACGIH TLV	OSHA PEL	NIOSH IDLH	Mexico OEL (TWA)
Potassium permanganate	TWA: 0.02 mg/m ³ TWA: 0.1 mg/m ³	(Vacated) Ceiling: 5 mg/m ³ Ceiling: 5 mg/m ³	IDLH: 500 mg/m ³ TWA: 1 mg/m ³ STEL: 3 mg/m ³	TWA: 0.2 mg/m ³

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH IDLH: NIOSH - National Institute for Occupational Safety and Health

Engineering Measures Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas.

Personal Protective Equipment

Eye/face Protection Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

Skin and body protection Wear appropriate protective gloves and clothing to prevent skin exposure.

Respiratory Protection Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

Hygiene Measures Handle in accordance with good industrial hygiene and safety practice.

9. Physical and chemical properties

Physical State	Solid Powder
Appearance	Dark brown
Odor	Odorless
Odor Threshold	No information available
pH	8 (16 g/l @ 20°C)
Melting Point/Range	240 °C / 464 °F
Boiling Point/Range	No information available

Flash Point	No information available
Evaporation Rate	Not applicable
Flammability (solid,gas)	No information available
Flammability or explosive limits	
Upper	No data available
Lower	No data available
Vapor Pressure	No information available
Vapor Density	Not applicable
Specific Gravity	2.700 g/cm3
Solubility	Soluble in water
Partition coefficient; n-octanol/water	No data available
Autoignition Temperature	Not applicable
Decomposition Temperature	240 °C
Viscosity	Not applicable
Molecular Formula	K Mn O4
Molecular Weight	158.04

10. Stability and reactivity

Reactive Hazard	Yes
Stability	Stable under normal conditions. Oxidizer: Contact with combustible/organic material may cause fire.
Conditions to Avoid	Incompatible products. Excess heat. Combustible material.
Incompatible Materials	Reducing Agent, Strong acids, Strong reducing agents, Combustible material
Hazardous Decomposition Products	Heavy metal oxides, Potassium oxides
Hazardous Polymerization	Hazardous polymerization does not occur.
Hazardous Reactions	None under normal processing.

11. Toxicological information

Acute Toxicity

Product Information

Oral LD50 Category 4.

Component Information

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Potassium permanganate	LD50 = 750 mg/kg (Rat)	LD50 > 2000 mg/kg (Rat)	Not listed

Toxicologically Synergistic Products No information available

Delayed and immediate effects as well as chronic effects from short and long-term exposure

Irritation	Causes severe irritation and or burns
Sensitization	No information available
Carcinogenicity	The table below indicates whether each agency has listed any ingredient as a carcinogen.

Component	CAS No	IARC	NTP	ACGIH	OSHA	Mexico
Potassium permanganate	7722-64-7	Not listed	Not listed	Not listed	Not listed	Not listed

Mutagenic Effects No information available

Reproductive Effects Possible risk of harm to the unborn child.

Developmental Effects	No information available.
Teratogenicity	No information available.
STOT - single exposure STOT - repeated exposure	Respiratory system Central nervous system (CNS) Brain
Aspiration hazard	No information available
Symptoms / effects, both acute and delayed	Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation
Endocrine Disruptor Information	No information available
Other Adverse Effects	The toxicological properties have not been fully investigated.

12. Ecological information

Ecotoxicity

The product contains following substances which are hazardous for the environment. Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

Component	Freshwater Algae	Freshwater Fish	Microtox	Water Flea
Potassium permanganate	0.41 mg/l/72h EC50	2.97-3.11 mg/L LC50 96 h 3.16-3.77 mg/L LC50 96 h 3.3-3.93 mg/L LC50 96 h 2.3 mg/L LC50 96 h 0.769-1.27 mg/L LC50 96 h 1.08-1.38 mg/L LC50 96 h 1.8-5.6 mg/L LC50 96 h 2.7 mg/L LC50 96 h	Not listed	Not listed

Persistence and Degradability May persist based on information available.

Bioaccumulation/ Accumulation No information available.

Mobility Will likely be mobile in the environment due to its water solubility.

Component	log Pow
Potassium permanganate	-1.73

13. Disposal considerations

Waste Disposal Methods Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

14. Transport information

DOT

UN-No UN1490
 Proper Shipping Name POTASSIUM PERMANGANATE
 Hazard Class 5.1
 Packing Group II

TDG

UN-No UN1490
 Proper Shipping Name POTASSIUM PERMANGANATE
 Hazard Class 5.1
 Packing Group II

IATA

UN-No UN1490

Proper Shipping Name	POTASSIUM PERMANGANATE
Hazard Class	5.1
Packing Group	II
IMDG/IMO	
UN-No	UN1490
Proper Shipping Name	POTASSIUM PERMANGANATE
Hazard Class	5.1
Packing Group	II

15. Regulatory information

United States of America Inventory

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	TSCA - EPA Regulatory Flags
Potassium permanganate	7722-64-7	X	ACTIVE	-

Legend:

TSCA US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

'-' - Not Listed

TSCA 12(b) - Notices of Export Not applicable

International Inventories

X = listed.

Component	CAS No	DSL	NDSL	EINECS	PICCS	ENCS	ISHL	AICS	IECSC	KECL
Potassium permanganate	7722-64-7	X	-	231-760-3	X	X	X	X	X	KE-29180

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

U.S. Federal Regulations

SARA 313

Component	CAS No	Weight %	SARA 313 - Threshold Values %
Potassium permanganate	7722-64-7	>95	1.0

SARA 311/312 Hazard Categories See section 2 for more information

CWA (Clean Water Act)

Component	CWA - Hazardous Substances	CWA - Reportable Quantities	CWA - Toxic Pollutants	CWA - Priority Pollutants
Potassium permanganate	X	100 lb	-	-

Clean Air Act

Component	HAPS Data	Class 1 Ozone Depletors	Class 2 Ozone Depletors
Potassium permanganate	X		-

OSHA - Occupational Safety and Health Administration Not applicable

CERCLA

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302)

Component	Hazardous Substances RQs	CERCLA EHS RQs
Potassium permanganate	100 lb	-

California Proposition 65

This product does not contain any Proposition 65 chemicals.

U.S. State Right-to-Know Regulations

Component	Massachusetts	New Jersey	Pennsylvania	Illinois	Rhode Island
Potassium permanganate	X	X	X	X	X

U.S. Department of Transportation

Reportable Quantity (RQ): Y
 DOT Marine Pollutant N
 DOT Severe Marine Pollutant N

U.S. Department of Homeland Security

This product contains the following DHS chemicals:

Legend - STQs = Screening Threshold Quantities, APA = A placarded amount

Component	DHS Chemical Facility Anti-Terrorism Standard
Potassium permanganate	Theft STQs - 400lb

Other International Regulations**Mexico - Grade**

No information available

Authorisation/Restrictions according to EU REACH

Component	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
Potassium permanganate	-	Use restricted. See item 75. (see link for restriction details)	-

<https://echa.europa.eu/substances-restricted-under-reach>
Safety, health and environmental regulations/legislation specific for the substance or mixture

Component	CAS No	OECD HPV	Persistent Organic Pollutant	Ozone Depletion Potential	Restriction of Hazardous Substances (RoHS)
Potassium permanganate	7722-64-7	Listed	Not applicable	Not applicable	Not applicable

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements	Rotterdam Convention (PIC)	Basel Convention (Hazardous Waste)
Potassium permanganate	7722-64-7	Not applicable	Not applicable	Not applicable	Not applicable

16. Other information**Prepared By**

Regulatory Affairs
 Thermo Fisher Scientific
 Email: EMSDS.RA@thermofisher.com

Creation Date

23-Oct-2009

Revision Date

24-Dec-2021

Print Date

24-Dec-2021

Revision Summary

This document has been updated to comply with the US OSHA HazCom 2012 Standard replacing the current legislation under 29 CFR 1910.1200 to align with the Globally Harmonized System of Classification and Labeling of Chemicals (GHS).

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

End of SDS

AIChE National Chem-E Car Competition		
ChemE-Car Engineering Design Package (EDP)		
Quality Assurance Evaluation		
Completed?	Design Parameter	Reviewer Comments/Noted
Technical Basis of Design		
Acceptable	Summary level description and diagram of car's design explaining its operation	
Acceptable	Discussion of power and stopping mechanism	
See Note	Identifies primary hazards present in car design	For maximum temperature do not state ambient. Please state the temperature of the exothermic reaction in the Thermoelectric energy generation
See Note	Includes LABELED pictures showing all sides of the car in a FULLY ASSEMBLED state.	please include picture of bottom of car and make sure all gears are sheilded.
N/A	If hydrogen is used, students are generating hydrogen ONSITE or ONBOARD the car, and below 5 psig (0.345 bar).	
Acceptable	Secondary containment is present for all parts of the car where liquid may be present.	
Acceptable	No manual addition of chemicals at the starting line is present (i.e. pouring or manual plunging of syringes).	
Acceptable	Caps and lids are secure. Foil, plastic wrap and parafilm are not permissible means of sealing a container.	
Acceptable	Electrical connections are shown. Connections must be secured and insulated. No alligator clips/twisted wires.	
Training and Certifications		
Acceptable	All team members have completed the AIChE Chem-E Car Safety Training Course (>80%) within the past 12 months.	
Acceptable	Certification sheet indicates students have operated the vehicle for at least 10 hrs.	
Acceptable	EDP Certification Page, signed by Faculty Supervisor and/or Consultant	
Hazard Analysis		
See Note	A legible and complete map of the team's laboratory and track area is provided, including egress and safety items.	Please label safety equipment in the map
Not Acceptable	Operating conditions (Press/Temp) are clearly defined (Max allowable Pressure = 200psig)	For maximum temperature do not state ambient. Please state the temperature of the
Acceptable	The types of PPE required is properly documented and matches the requirements of the JSA.	
Acceptable	Vehicle primary Hazard Checklist is complete (i.e. Pressure, Toxicity, Flammable, Temperature...)	
Acceptable	Fabrication & Operation Hazard Detail Check List is complete, lists hazards, control methods & PPE.	
Acceptable	Description of chemistry involved and chemical reactions including heat generation and side products is shown.	
Acceptable	Chemical information page is complete and shows the amount of each INDIVIDUAL CHEMICAL required.	
Not Acceptable	Chemical Hazards and Disposal information is complete and lists waste disposal classification for each substance.	Please fill out waste forms

ChemE-Car Engineering Design Package (EDP) Quality Assurance Evaluation		
Completed?	Design Parameter	Reviewer Comments/Noted
Not Acceptable	Team Waste Tags are completely filled out for each anticipated waste stream (volume may be blank) and are included in the EDP.	Please fill out waste forms
Acceptable	Safety Data Sheets (SDS) are provided for all chemicals and waste products and included in the EDP Supplement.	
Operating Procedures		
Acceptable	Detailed and appropriate operating procedures are provided for all operations, including preparation of chemical solutions.	
Acceptable	All chemical handling and waste disposal procedures are included and align with the JSA.	
Acceptable	Operations between runs and cleanup at the end of the competition are documented	
Acceptable	Procedures include steps for disassembling all student built electro chemical (battery) devices and making the individual components safe prior to disposal.	
Equipment Design Basis		
Acceptable	Tabulated and complete list of every piece of the car, including the maximum design P/T for each component.	
Acceptable	Specification/data sheets for purchased equipment from the manufacturer are included in the EDP Supplement	
N/A	LEL calculations are provided for any hydrogen discharge from fuel cell cars	
N/A	Any gaseous discharge/venting from the car has an NFPA rating less than 3,	
Pressure Relief Design Basis		
N/A	Pressure vessel has certification from manufacturer or students have provided testing procedure and results of the testing per Appendix in the Safety Rules.	
N/A	No pressurized plastic components are allowed on the car even if it has a pressure rating. No exceptions.	
N/A	Quantitative design basis calculations for the pressure relieving load requirements (students must explain how they determined the pressure relief requirements using calculations).	
N/A	Sizing calculations for the pressure relief device that meets expected load (students must show calculations which demonstrate that the PRV orifice size/capacity is appropriate for the needed flowrate)	
N/A	The relief device setpoint is fixed and stamped or if variable has been tested per the Safety Rules, witnessed, and signed off by a faculty advisor or non-student expert.	
N/A	The relief device is installed directly on the source of pressure generation without any restrictions or valves between it and the pressure vessel.	



MANAGEMENT OF CHANGE REQUEST FORM

Name of Person or Group Requesting Change: Sharkus Electrus

Date: 04/25/2025

Summary Description of Change:

- We are not using CaO anymore

Reason(s) for Change:

- We did not need to have a very strong exothermic reaction

Change is: ☒ permanent
☐ temporary Duration of change: _____

Signature of Requestor: Gustavo De la Torre

	Yes	N/A
Attachment checklist:	☐	☒ Updated Process Flow Sheet, if changed
	☐	☒ Current Process Flow Sheet
	☐	☒ Updated JSA pages, if changed
	☐	☒ Current JSA pages
	☒	☐ Updated Engineering documentation pages, if changed
	☒	☐ Current Engineering documentation pages
	☒	Safety Evaluation Checklist (req'd. – see below)

Management of Change Request Form

PROCESS CHANGE REQUEST APPROVAL FORM

Type of Change:

- ☒ Major – Requires review by entire team and faculty adviser
☐ Minor – Requires review by team only

Description of Change as Approved:

•

Approved Duration of Change: Ji-Hwan Kang

Signatures of Approval

Faculty Adviser: Ji-Hwan Kang Date: 04/25/2025

Review Team Member: Adrian Zentento Date: 04/25/2025

Review Team Member: Alejandro Jimenez Date: 04/25/2025

Review Team Member: Jason Aguilar Date: 04/25/2025

Review Team Member: Wailin Thai Date: 04/25/2025

Review Team Member: Mathew Tan Date: 04/25/2025

Review Team Member: Taylor Soop Date: 04/25/2025

Attach minutes and pertinent notes from review team meetings and place in permanent file.

Management of Change Request Form**SAFETY EVALUATION CHECKLIST**

Complete this safety evaluation checklist, noting any conditions that apply to the proposed change. Attach a more detailed description of the change, if necessary, along with all supporting materials.

Changes in Process Conditions/Materials:

- ☒ temperature
- ☐ pressure
- ☐ flow
- ☐ level
- ☐ composition
- ☒ chemical
- ☐ toxicity
- ☐ flash point
- ☐ reaction conditions
- ☐ biohazard
- ☐ use of regulated materials
- ☐ use of lasers/ change in class of lasers

Changes in Operating Conditions/ Procedures:

- ☐ startup
- ☒ normal operations
- ☐ abnormal operations
- ☐ shut down
- ☐ emergency shut down
- ☐ maintenance procedures
- ☐ lock-out
- ☐ tagging

Changes in Process Equipment:

- ☒ size or capacity
- ☐ materials of construction
- ☐ seals and gasket materials
- ☐ piping/ valving
- ☐ electrical system
- ☐ max./min. working pressure
- ☒ max./min. working temperature
- ☐ equipment guarding
- ☐ thermal insulation/insulation coverings
- ☐ filters
- ☐ support structures

Changes in Facilities:

- ☐ ventilation
- ☐ lighting
- ☐ utility services
- ☐ building modifications/renovations

Changes in Safety Equipment:

- ☐ containment
- ☐ guarding
- ☐ fire protection
- ☐ fire detection
- ☐ safety equipment location
- ☐ safety equipment type
- ☐ personal protective equipment
- ☐ grounding/bonding

Changes in Control Systems/ Elements:

- ☐ programming change
- ☐ controller action
- ☐ control valve trim/sizing
- ☐ instrument/transmitter
- ☐ sensing element
- ☐ graphically displayed information
- ☐ system response to alarm

Changes in Environmental Conditions:

- ☐ discharge air quality
- ☐ discharge water quality
- ☐ solid waste



MANAGEMENT OF CHANGE REQUEST FORM

Name of Person or Group Requesting Change: Sharkus Electrus

Date: 04/25/2025

Summary Description of Change:

- Changed the amount of Thermo-Electric Generators from 6 to 8 separated into two cells

Reason(s) for Change:

- Needed more power output

Change is: ☒ permanent
☐ temporary Duration of change: _____

Signature of Requestor: Gustavo De la Torre

	Yes	N/A
Attachment checklist:	☐	☒ Updated Process Flow Sheet, if changed
	☐	☒ Current Process Flow Sheet
	☐	☒ Updated JSA pages, if changed
	☐	☒ Current JSA pages
	☒	☐ Updated Engineering documentation pages, if changed
	☒	☐ Current Engineering documentation pages
	☒	Safety Evaluation Checklist (req'd. – see below)

Management of Change Request Form

PROCESS CHANGE REQUEST APPROVAL FORM

Type of Change: ☒ Major – Requires review by entire team and faculty adviser
☐ Minor – Requires review by team only

Description of Change as Approved:

-

Approved Duration of Change: Ji-Hwan Kang

Signatures of Approval

Faculty Adviser: Ji-Hwan Kang Date: 04/25/2025

Review Team Member: Adrian Zentento Date: 04/25/2025

Review Team Member: Alejandro Jimenez Date: 04/25/2025

Review Team Member: Jason Aguilar Date: 04/25/2025

Review Team Member: Wailin Thai Date: 04/25/2025

Review Team Member: Mathew Tan Date: 04/25/2025

Review Team Member: Taylor Soop Date: 04/25/2025

Attach minutes and pertinent notes from review team meetings and place in permanent file.

Management of Change Request Form**SAFETY EVALUATION CHECKLIST**

Complete this safety evaluation checklist, noting any conditions that apply to the proposed change. Attach a more detailed description of the change, if necessary, along with all supporting materials.

Changes in Process Conditions/Materials:

- ☐ temperature
- ☐ pressure
- ☐ flow
- ☐ level
- ☐ composition
- ☐ chemical
- ☐ toxicity
- ☐ flash point
- ☐ reaction conditions
- ☐ biohazard
- ☐ use of regulated materials
- ☐ use of lasers/ change in class of lasers

Changes in Operating Conditions/ Procedures:

- ☐ startup
- ☐ normal operations
- ☐ abnormal operations
- ☐ shut down
- ☐ emergency shut down
- ☐ maintenance procedures
- ☐ lock-out
- ☐ tagging

Changes in Process Equipment:

- ☒ size or capacity
- ☒ materials of construction
- ☐ seals and gasket materials
- ☐ piping/ valving
- ☒ electrical system
- ☐ max./min. working pressure
- ☐ max./min. working temperature
- ☐ equipment guarding
- ☐ thermal insulation/insulation coverings
- ☐ filters
- ☐ support structures

Changes in Facilities:

- ☐ ventilation
- ☐ lighting
- ☐ utility services
- ☒ building modifications/renovations

Changes in Safety Equipment:

- ☐ containment
- ☐ guarding
- ☐ fire protection
- ☐ fire detection
- ☐ safety equipment location
- ☐ safety equipment type
- ☐ personal protective equipment
- ☐ grounding/bonding

Changes in Control Systems/ Elements:

- ☐ programming change
- ☐ controller action
- ☐ control valve trim/sizing
- ☐ instrument/transmitter
- ☐ sensing element
- ☐ graphically displayed information
- ☐ system response to alarm

Changes in Environmental Conditions:

- ☐ discharge air quality
- ☐ discharge water quality
- ☐ solid waste



MANAGEMENT OF CHANGE REQUEST FORM

Name of Person or Group Requesting Change: Sharkus Electrus

Date: 04/25/2025

Summary Description of Change:

- Added a taller container for the Dry Ice and the solution containers
- Added a new container on top of the solution containers

Reason(s) for Change:

- Safer to contain the Dry Ice and the solutions
- Easier to pump the HCl into the NaOH

Change is: ☒ permanent
☐ temporary Duration of change: _____

Signature of Requestor: Gustavo De la Torre

	Yes	N/A
Attachment checklist:	☐	☒ Updated Process Flow Sheet, if changed
	☐	☒ Current Process Flow Sheet
	☐	☒ Updated JSA pages, if changed
	☐	☒ Current JSA pages
	☒	☐ Updated Engineering documentation pages, if changed
	☒	☐ Current Engineering documentation pages
	☒	Safety Evaluation Checklist (req'd. – see below)

Management of Change Request Form

PROCESS CHANGE REQUEST APPROVAL FORM

Type of Change: ☒ Major – Requires review by entire team and faculty adviser
☐ Minor – Requires review by team only

Description of Change as Approved:

-

Approved Duration of Change: Ji-Hwan Kang

Signatures of Approval

Faculty Adviser: Ji-Hwan Kang Date: 04/25/2025

Review Team Member: Adrian Zentento Date: 04/25/2025

Review Team Member: Alejandro Jimenez Date: 04/25/2025

Review Team Member: Jason Aguilar Date: 04/25/2025

Review Team Member: Wailin Thai Date: 04/25/2025

Review Team Member: Mathew Tan Date: 04/25/2025

Review Team Member: Taylor Soop Date: 04/25/2025

Attach minutes and pertinent notes from review team meetings and place in permanent file.

Management of Change Request Form**SAFETY EVALUATION CHECKLIST**

Complete this safety evaluation checklist, noting any conditions that apply to the proposed change. Attach a more detailed description of the change, if necessary, along with all supporting materials.

Changes in Process Conditions/Materials:

- ☐ temperature
- ☐ pressure
- ☐ flow
- ☒ level
- ☐ composition
- ☐ chemical
- ☐ toxicity
- ☐ flash point
- ☐ reaction conditions
- ☐ biohazard
- ☐ use of regulated materials
- ☐ use of lasers/ change in class of lasers

Changes in Operating Conditions/ Procedures:

- ☐ startup
- ☐ normal operations
- ☐ abnormal operations
- ☐ shut down
- ☐ emergency shut down
- ☐ maintenance procedures
- ☐ lock-out
- ☐ tagging

Changes in Process Equipment:

- ☒ size or capacity
- ☒ materials of construction
- ☒ seals and gasket materials
- ☐ piping/ valving
- ☐ electrical system
- ☐ max./min. working pressure
- ☐ max./min. working temperature
- ☒ equipment guarding
- ☒ thermal insulation/insulation coverings
- ☐ filters
- ☒ support structures

Changes in Facilities:

- ☐ ventilation
- ☐ lighting
- ☐ utility services
- ☒ building modifications/renovations

Changes in Safety Equipment:

- ☒ containment
- ☒ guarding
- ☐ fire protection
- ☐ fire detection
- ☐ safety equipment location
- ☐ safety equipment type
- ☐ personal protective equipment
- ☐ grounding/bonding

Changes in Control Systems/ Elements:

- ☐ programming change
- ☐ controller action
- ☐ control valve trim/sizing
- ☐ instrument/transmitter
- ☐ sensing element
- ☐ graphically displayed information
- ☐ system response to alarm

Changes in Environmental Conditions:

- ☐ discharge air quality
- ☐ discharge water quality
- ☐ solid waste



MANAGEMENT OF CHANGE REQUEST FORM

Name of Person or Group Requesting Change: Sharkus Electrus

Date: 04/25/2025

Summary Description of Change:

- We added pumps to each container of the solution and a container on top of both to circulate the HCl initially.

Reason(s) for Change:

- We needed constant circulation of the solution

Change is: ☒ permanent
☐ temporary Duration of change: _____

Signature of Requestor: Gustavo De la Torre

	Yes	N/A
Attachment checklist:	☐	☒ Updated Process Flow Sheet, if changed
	☐	☒ Current Process Flow Sheet
	☐	☒ Updated JSA pages, if changed
	☐	☒ Current JSA pages
	☒	☐ Updated Engineering documentation pages, if changed
	☒	☐ Current Engineering documentation pages
	☒	Safety Evaluation Checklist (req'd. – see below)

Management of Change Request Form

PROCESS CHANGE REQUEST APPROVAL FORM

Type of Change: ☒ Major – Requires review by entire team and faculty adviser
☐ Minor – Requires review by team only

Description of Change as Approved:

-

Approved Duration of Change: Ji-Hwan Kang

Signatures of Approval

Faculty Adviser: Ji-Hwan Kang Date: 04/25/2025

Review Team Member: Adrian Zentento Date: 04/25/2025

Review Team Member: Alejandro Jimenez Date: 04/25/2025

Review Team Member: Jason Aguilar Date: 04/25/2025

Review Team Member: Wailin Thai Date: 04/25/2025

Review Team Member: Mathew Tan Date: 04/25/2025

Review Team Member: Taylor Soop Date: 04/25/2025

Attach minutes and pertinent notes from review team meetings and place in permanent file.

Management of Change Request Form**SAFETY EVALUATION CHECKLIST**

Complete this safety evaluation checklist, noting any conditions that apply to the proposed change. Attach a more detailed description of the change, if necessary, along with all supporting materials.

Changes in Process Conditions/Materials:

- ☐ temperature
- ☐ pressure
- ☒ flow
- ☐ level
- ☐ composition
- ☐ chemical
- ☐ toxicity
- ☐ flash point
- ☐ reaction conditions
- ☐ biohazard
- ☐ use of regulated materials
- ☐ use of lasers/ change in class of lasers

Changes in Operating Conditions/ Procedures:

- ☐ startup
- ☐ normal operations
- ☐ abnormal operations
- ☐ shut down
- ☐ emergency shut down
- ☐ maintenance procedures
- ☐ lock-out
- ☐ tagging

Changes in Process Equipment:

- ☐ size or capacity
- ☒ materials of construction
- ☒ seals and gasket materials
- ☒ piping/ valving
- ☐ electrical system
- ☐ max./min. working pressure
- ☒ max./min. working temperature
- ☐ equipment guarding
- ☐ thermal insulation/insulation coverings
- ☐ filters
- ☐ support structures

Changes in Facilities:

- ☐ ventilation
- ☐ lighting
- ☐ utility services
- ☒ building modifications/renovations

Changes in Safety Equipment:

- ☐ containment
- ☐ guarding
- ☐ fire protection
- ☐ fire detection
- ☐ safety equipment location
- ☐ safety equipment type
- ☐ personal protective equipment
- ☐ grounding/bonding

Changes in Control Systems/ Elements:

- ☐ programming change
- ☐ controller action
- ☐ control valve trim/sizing
- ☐ instrument/transmitter
- ☐ sensing element
- ☐ graphically displayed information
- ☐ system response to alarm

Changes in Environmental Conditions:

- ☐ discharge air quality
- ☐ discharge water quality
- ☐ solid waste



MANAGEMENT OF CHANGE REQUEST FORM

Name of Person or Group Requesting Change: Sharkus Electrus

Date: 04/25/2025

Summary Description of Change:

- Added a relay connecting the Arduino board and the motor

Reason(s) for Change:

- Needed a relay so we could control the voltage going to the motor and so we would not burn the Arduino board with our battery's voltage

Change is: ☒ permanent
☐ temporary Duration of change: _____

Signature of Requestor: Gustavo De la Torre

	Yes	N/A
Attachment checklist:	☐	☒ Updated Process Flow Sheet, if changed
	☐	☒ Current Process Flow Sheet
	☐	☒ Updated JSA pages, if changed
	☐	☒ Current JSA pages
	☒	☐ Updated Engineering documentation pages, if changed
	☒	☐ Current Engineering documentation pages
	☒	Safety Evaluation Checklist (req'd. – see below)

Management of Change Request Form

PROCESS CHANGE REQUEST APPROVAL FORM

Type of Change: ☒ Major – Requires review by entire team and faculty adviser
☐ Minor – Requires review by team only

Description of Change as Approved:

-

Approved Duration of Change: Ji-Hwan Kang

Signatures of Approval

Faculty Adviser: Ji-Hwan Kang Date: 04/25/2025

Review Team Member: Adrian Zentento Date: 04/25/2025

Review Team Member: Alejandro Jimenez Date: 04/25/2025

Review Team Member: Jason Aguilar Date: 04/25/2025

Review Team Member: Wailin Thai Date: 04/25/2025

Review Team Member: Mathew Tan Date: 04/25/2025

Review Team Member: Taylor Soop Date: 04/25/2025

Attach minutes and pertinent notes from review team meetings and place in permanent file.

Management of Change Request Form**SAFETY EVALUATION CHECKLIST**

Complete this safety evaluation checklist, noting any conditions that apply to the proposed change. Attach a more detailed description of the change, if necessary, along with all supporting materials.

Changes in Process Conditions/Materials:

- ☐ temperature
- ☐ pressure
- ☐ flow
- ☐ level
- ☐ composition
- ☐ chemical
- ☐ toxicity
- ☐ flash point
- ☐ reaction conditions
- ☐ biohazard
- ☐ use of regulated materials
- ☐ use of lasers/ change in class of lasers

Changes in Operating Conditions/ Procedures:

- ☒ startup
- ☐ normal operations
- ☐ abnormal operations
- ☒ shut down
- ☐ emergency shut down
- ☐ maintenance procedures
- ☐ lock-out
- ☐ tagging

Changes in Process Equipment:

- ☐ size or capacity
- ☒ materials of construction
- ☐ seals and gasket materials
- ☐ piping/ valving
- ☒ electrical system
- ☐ max./min. working pressure
- ☐ max./min. working temperature
- ☐ equipment guarding
- ☐ thermal insulation/insulation coverings
- ☐ filters
- ☐ support structures

Changes in Facilities:

- ☐ ventilation
- ☐ lighting
- ☐ utility services
- ☒ building modifications/renovations

Changes in Safety Equipment:

- ☐ containment
- ☐ guarding
- ☐ fire protection
- ☐ fire detection
- ☐ safety equipment location
- ☐ safety equipment type
- ☐ personal protective equipment
- ☐ grounding/bonding

Changes in Control Systems/ Elements:

- ☒ programming change
- ☒ controller action
- ☐ control valve trim/sizing
- ☒ instrument/transmitter
- ☐ sensing element
- ☐ graphically displayed information
- ☐ system response to alarm

Changes in Environmental Conditions:

- ☐ discharge air quality
- ☐ discharge water quality
- ☐ solid waste



MANAGEMENT OF CHANGE REQUEST FORM

Name of Person or Group Requesting Change: Sharkus Electrus

Date: 04/25/2025

Summary Description of Change:

- Added 3 switches

Reason(s) for Change:

- One to reset the Arduino
- Another to turn on the pumps
- And the last one to start the motor

Change is: ☒ permanent
☐ temporary Duration of change: _____

Signature of Requestor: Gustavo De la Torre

	Yes	N/A
Attachment checklist:	☐	☒ Updated Process Flow Sheet, if changed
	☐	☒ Current Process Flow Sheet
	☐	☒ Updated JSA pages, if changed
	☐	☒ Current JSA pages
	☒	☐ Updated Engineering documentation pages, if changed
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Management of Change Request Form

PROCESS CHANGE REQUEST APPROVAL FORM

Type of Change: ☒ Major – Requires review by entire team and faculty adviser
☐ Minor – Requires review by team only

Description of Change as Approved:

-

Approved Duration of Change: Ji-Hwan Kang

Signatures of Approval

Faculty Adviser: Ji-Hwan Kang Date: 04/25/2025

Review Team Member: Adrian Zentento Date: 04/25/2025

Review Team Member: Alejandro Jimenez Date: 04/25/2025

Review Team Member: Jason Aguilar Date: 04/25/2025

Review Team Member: Wailin Thai Date: 04/25/2025

Review Team Member: Mathew Tan Date: 04/25/2025

Review Team Member: Taylor Soop Date: 04/25/2025

Attach minutes and pertinent notes from review team meetings and place in permanent file.

Management of Change Request Form

SAFETY EVALUATION CHECKLIST

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- ☐ level
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- ☐ chemical
- ☐ toxicity
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- ☐ reaction conditions
- ☐ biohazard
- ☐ use of regulated materials
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Changes in Operating Conditions/ Procedures:

- ☒ startup
- ☐ normal operations
- ☐ abnormal operations
- ☒ shut down
- ☒ emergency shut down
- ☐ maintenance procedures
- ☐ lock-out
- ☐ tagging

Changes in Process Equipment:

- ☐ size or capacity
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- ☐ filters
- ☐ support structures

Changes in Facilities:

- ☐ ventilation
- ☐ lighting
- ☐ utility services
- ☒ building modifications/renovations

Changes in Safety Equipment:

- ☐ containment
- ☐ guarding
- ☐ fire protection
- ☐ fire detection
- ☐ safety equipment location
- ☐ safety equipment type
- ☐ personal protective equipment
- ☐ grounding/bonding

Changes in Control Systems/ Elements:

- ☒ programming change
- ☒ controller action
- ☐ control valve trim/sizing
- ☒ instrument/transmitter
- ☐ sensing element
- ☐ graphically displayed information
- ☐ system response to alarm

Changes in Environmental Conditions:

- ☐ discharge air quality
- ☐ discharge water quality
- ☐ solid waste